

L-convex polyominoes and a colored-partition comparison

A proof of the A126764 asymptotic, three correction terms,
an all-orders radial expansion, and inverse asymptotics

Part II: the all-orders coefficient theorem

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Abstract

Let a_n count fixed L-convex polyominoes of area n , with $a_0 = 1$. Starting from the established area generating function, we prove the Guttman–Kotěšovec asymptotic recorded as a conjecture in OEIS A126764:

$$a_n \sim \frac{13\sqrt{2}}{768 n^{3/2}} \exp\left(\pi\sqrt{\frac{13n}{6}}\right).$$

The main device is a positive comparison with a colored-partition Euler product $R(q)$, given explicitly below. If $p_n = [q^n]R(q)$, we obtain the finite, coefficientwise bound $0 \leq p_n - 4a_n \leq 3(p_n - 2p_{n-1} + p_{n-2})$ for every $n \geq 3$. A higher finite-cut identity then proves three explicit correction terms, with relative remainder $O(n^{-7/4})$. Independently, a finite recurrence computes the complete radial Poincaré expansion of $\mathcal{A}(e^{-t})$ to any prescribed order. We also derive an asymptotic inverse, including the necessary distinction between a real interpolation and an integer threshold. Exact integer checks through $n = 2000$ and symbolic coefficient checks accompany the manuscript. No numerical fit enters the proofs.

Scope and status. The proofs below are conventional mathematical proofs, not Lean certificates or a claim of independent peer review. The exact enumerative generating function is a cited input; its asymptotic analysis is supplied here. The all-orders *radial* expansion is proved. An all-orders *coefficient* expansion, and exponentially small transseries sectors for a_n , are not claimed.

[write] Added 2 October 2026, batch 77P5. The box describes Part I. Part II (Sections 13–23), a second manuscript, proves the all-orders *coefficient* expansion (Theorem 14.1). Exponentially small transseries sectors remain unclaimed in both parts.

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1 The problem and the main results

1.1 The sequence and the source boundary

An L-convex polyomino is a convex polyomino in which any two cells can be joined inside the polyomino by a path with at most one turn. We count fixed polyominoes: translations are identified, but rotations and reflections are not generally identified. This is the convention in OEIS A126764 [1]. The initial terms are

$$1, 1, 2, 6, 15, 35, 76, 156, 310, 590, 1098, 1984, 3515, 6094, \dots$$

The zero-area term is a generating-function convention.

Castiglione, Frosini, Munarini, Restivo, and Rinaldi established an area generating function [2]; we use its explicit recurrence form as reproduced in Guttmann and Kotěšovec [3], equation (1.1). The latter authors proposed the leading asymptotic from a numerical analysis. The OEIS entry inspected for this report still labels that formula a conjecture [1]. A targeted search did not locate an earlier proof of the specific area asymptotic. This search is not a comprehensive priority certification.

The ProveIt repository [7] was inspected as the proposed research context. A targeted repository search for A126764 returned no matching file. The present manuscript is a standalone contribution: it does not assume any unverified theorem from another repository report. Enumeration by perimeter, and the asymptotics of 201-avoiding ascent sequences discussed alongside L-convex polyominoes in [3], are separate questions and are not asserted to be solved here.

[write] Added 2 October 2026, batch 75. This report is a single manuscript, printed in full: manuscript 05 of batch 75 of ProveIt’s incoming-reports intake, delivered as the archive A126764_Lconvex_Research.zip in commit 4b874cea0 (main file lconvex_asymptotics.tex, now article.tex) and placed in commit 6ea60e367 in the research-report collection, among the OEIS-sequence asymptotics reports. The manuscript names no ProveIt commit: its repository input was an overview and a targeted search for A126764 on 1 October 2026, so it has no pin. That search is still accurate for the repository it saw; the only match now is this report. Its PDF metadata records it as “prepared with ChatGPT”. Nothing in this report is formalized in Lean or Rocq, and placement in the collection confers no formal status on any statement here. The writing step changed only the labels (every label now carries the prefix lcp:), set the bibliography ragged-right, and added these [write] notes and two references [8, 9]; no statement, proof or number of the manuscript was altered. The polyomino report nearest in subject, polyomino-growth-finite-prefix-corrections [9], bounds Klarner’s growth constant $\lambda \leq 4.498$ for *all* polyominoes (exponential growth, with a Lean development Combinatorics/Polyominoes/KlarnerConstant for the weaker bound 9047/2000); the L-convex class grows only like $e^{B\sqrt{n}}$, so the two constants are unrelated and no theorem is shared. The report skew-partition-continued-fractions touches parallelogram polyominoes (A006958) only.

[write] Added 2 October 2026, batch 77P1. The 201-avoiding ascent sequences that the preceding paragraph sets aside are now treated in the sibling report a202062-ascent-201-enumeration (manuscript 43 of batch 77): it proves the Guttmann–Kotěšovec algebraic (cubic) generating function for A202062 and its all-orders coefficient asymptotics. The two problems share only the paper [3]; no theorem, constant or method of this report is used there, and nothing here changes.

[write] Added 2 October 2026, batch 77P5. The note above describes Part I, Sections 1–12 and Appendices A and B. Since batch 77 the report has a second part, Part II (Sections 13–23), printed in full from manuscript 31 of batch 77 (archive lconvex-asymptotics-reproducibility.zip, arrival commit 096ee7b87, placement commit 4f11bc9c0; no pin). It proves the all-orders coefficient expansion that Part I leaves open, by an exact decomposition $\mathcal{A} = R D^2 +$ (an exponentially smaller remainder) in Part I’s letters. Its letters clash with Part I’s; Table 1 resolves them.

1.2 Constants and coefficient asymptotics

Throughout, put

$$K = \frac{13\pi^2}{24}, \quad B = 2\sqrt{K} = \pi\sqrt{\frac{13}{6}}, \quad c = \frac{13\sqrt{2}}{768}. \quad (1.1)$$

The letter K denotes a constant, not a generating function.

Theorem 1.1 (Coefficient expansion). *As $n \rightarrow \infty$ through the positive integers,*

$$a_n = c n^{-3/2} e^{B\sqrt{n}} \left(1 - \frac{d}{\sqrt{n}} + \frac{e}{n} + \frac{f}{n^{3/2}} + O(n^{-7/4}) \right), \quad (1.2)$$

where

$$d = \frac{\sqrt{K}}{6} + \frac{3}{2\sqrt{K}} = \frac{B}{12} + \frac{3}{B}, \quad (1.3)$$

$$e = \frac{1}{2} + \frac{3}{4K} - \frac{35K}{72}, \quad (1.4)$$

$$f = \frac{755K^{3/2}}{1296} + \frac{175\sqrt{K}}{72} - \frac{5}{8\sqrt{K}}. \quad (1.5)$$

Numerically,

$$d = 1.0341052176262747\dots, \quad e = -1.9584764928967362\dots, \quad f = 12.550445779146218\dots$$

In particular, the leading asymptotic stated in the abstract holds.

There are two complementary proof routes. The first proves the leading formula and its $n^{-1/2}$ correction by an explicit positive squeeze. The second uses a finite rational approximation and a contour estimate; it proves all the terms in Theorem 1.1.

Theorem 1.2 (A finite coefficient squeeze). *Define*

$$R(q) = \frac{(-q; q^2)_\infty}{(q; q)_\infty^3} = \frac{(q^2; q^2)_\infty^2}{(q; q)_\infty^4 (q^4; q^4)_\infty} = \sum_{n \geq 0} p_n q^n, \quad J(q) = (1 - q)^2 R(q) = \sum_{n \geq 0} j_n q^n. \quad (1.6)$$

Then, for every $n \geq 3$,

$$0 \leq p_n - 4a_n \leq 3j_n = 3(p_n - 2p_{n-1} + p_{n-2}). \quad (1.7)$$

Both R and J have nonnegative integer coefficients. Moreover,

$$n \left(1 - \frac{4a_n}{p_n} \right) \longrightarrow \frac{K}{2} = \frac{13\pi^2}{48}. \quad (1.8)$$

The product R counts partitions with four colors available for an odd part, two for a part congruent to 2 modulo 4, and three for a multiple of 4. Thus the constant 13 comes from the average number $13/4$ of colors, not from a fit to a decimal growth constant. The factor $1/4$ in the comparison is forced by two boundary conditions of the recurrence. We do not assert a four-to-one combinatorial map; finding one is a natural further problem.

1.3 What the all-orders statement means

Let $t > 0$ and $q = e^{-t}$. We will prove

$$\mathcal{A}(e^{-t}) \sim \frac{t^{3/2}}{4(2\pi)^{3/2}} \exp\left(\frac{K}{t} - \frac{t}{6}\right) \mathcal{F}(t), \quad (1.9)$$

where $\mathcal{F}$ is a formally specified series beginning

$$\begin{aligned} \mathcal{F}(t) = & 1 - \frac{1}{2}t^2 + \frac{1}{2}t^3 - \frac{41}{48}t^4 + \frac{7}{4}t^5 - \frac{6317}{1440}t^6 \\ & + \frac{1529}{120}t^7 - \frac{1702513}{40320}t^8 + \dots \end{aligned} \quad (1.10)$$

Here $\sim$ is meant to all fixed algebraic orders along the positive real axis. It is not a claim that this series converges, nor that a real-axis expansion alone implies coefficient asymptotics. The complex estimate needed for the three coefficient corrections is proved separately.

2 An exact recurrence with a positive normalization

Write $(a; q)_m = \prod_{j=0}^{m-1} (1 - aq^j)$, and set $(q; q)_0 = 1$. The enumerative input is

$$\mathcal{A}(q) = 1 + \sum_{n \geq 1} \frac{q^n h_n(q)}{(q; q)_{n-1}^2 (1 - q^n)}, \quad (2.1)$$

where

$$h_0 = 1, \quad h_1 = 1, \quad h_{n+1} = 2h_n - (1 - q^n)^2 h_{n-1} \quad (n \geq 1). \quad (2.2)$$

Indeed, $h_n = f_{n-1}$ for $n \geq 1$ in the notation of [3], and $h_2 = 1 + 2q - q^2$. Introducing h_0 makes the recurrence valid already at $n = 1$.

The auxiliary solution

$$Q_0 = 1, \quad Q_1 = 2, \quad Q_{n+1} = 2Q_n - (1 - q^n)^2 Q_{n-1} \quad (2.3)$$

will produce the comparison product. For a general solution y_n of the same recurrence, define

$$b_n = \frac{y_n}{(q; q)_n^2}, \quad e_n = b_n - b_{n-1}, \quad T_n(q) = \frac{1}{(1 - q^n)^2} - 1.$$

We reserve d, e, f without subscripts for the constants in Theorem 1.1.

Lemma 2.1 (Positive transfer). *For $n \geq 2$,*

$$\begin{aligned} b_n &= \frac{b_{n-1} + e_{n-1}}{(1 - q^n)^2}, \\ e_n &= e_{n-1} + T_n(q)(b_{n-1} + e_{n-1}). \end{aligned} \quad (2.4)$$

Every entry of this linear transfer rule is a power series with nonnegative coefficients. Consequently, coefficientwise inequalities between two pairs (b_m, e_m) propagate to all later pairs and to their limiting slopes, whenever the slopes exist.

Proof. Dividing the recurrence by $(q; q)_{n-1}^2$ gives $(1 - q^n)^2 b_n = 2b_{n-1} - b_{n-2} = b_{n-1} + e_{n-1}$. Subtract b_{n-1} . Finally, $T_n = \sum_{r \geq 1} (r+1)q^{nr}$ has nonnegative coefficients. $\square$

For h , the initial normalized pair is

$$b_1 = (1 - q)^{-2}, \quad e_1 = (1 - q)^{-2} - 1,$$

and both entries are nonnegative. The same holds for Q , with $b_1 = 2(1 - q)^{-2}$ and $e_1 = 2(1 - q)^{-2} - 1$. Throughout, $F \preceq G$ means that every coefficient of $G - F$ is nonnegative. This is a stronger relation than $F(q) \leq G(q)$ for $0 < q < 1$. We will be careful not to multiply a coefficientwise inequality by a polynomial with negative coefficients.

3 Limiting slopes and the exact Euler product

3.1 Existence and analyticity of the slopes

Lemma 3.1. *For a solution with polynomial initial values and every $0 < r < 1$, there are analytic functions $s(q), b(q)$ on $|q| < 1$ such that, uniformly on $|q| \leq r$,*

$$y_n(q) = s(q)n + b(q) + O_r(n^2 r^n).$$

In particular, the normalized limiting slope

$$\mathcal{L}_y(q) = \lim_{n \rightarrow \infty} \left(\frac{y_n}{(q; q)_n^2} - \frac{y_{n-1}}{(q; q)_{n-1}^2} \right) = \frac{s(q)}{(q; q)_\infty^2} \quad (3.1)$$

exists locally uniformly. The same limit exists coefficientwise as a formal power series.

Proof. Put $v_j = 2q^j - q^{2j}$. Twice summing $y_{j+1} - 2y_j + y_{j-1} = v_j y_{j-1}$ gives

$$y_n = y_0 + n(y_1 - y_0) + \sum_{j=1}^{n-1} (n-j)v_j y_{j-1}.$$

On $|q| \leq r$, $|v_j| \leq 3r^j$. Divide by $n+1$ and apply the elementary finite-product version of discrete Gronwall's inequality. The resulting bound is a constant times $\prod_{j \geq 1} (1 + 3jr^j) < \infty$. Thus $y_n = O_r(n+1)$. The increments $y_{n+1} - y_n$ have a uniformly convergent sum of successive differences, with exponentially small tails. One more summation yields the displayed affine expansion. Since $(q; q)_n / (q; q)_\infty = 1 + O_r(r^n)$, normalization gives (3.1). Finally, in (2.4) the change in e_n has valuation at least n . Each fixed coefficient therefore stabilizes. $\square$

Write $L = \mathcal{L}_h$. If

$$C(z; q) = \sum_{n \geq 0} \frac{h_n(q)}{(q; q)_n^2} z^n, \quad C_j(q) = C(q^j; q), \quad (3.2)$$

then C has radius of convergence at least 1 in z and a possible double pole at $z = 1$ with leading coefficient L . Equation (2.1) becomes

$$\mathcal{A} = 1 + C_1 - C_2. \quad (3.3)$$

Multiplying the normalized recurrence by z^n and summing, including the initial condition $h_1 = 1$, yields

$$(1 - z)^2 C(z; q) - 2C(qz; q) + C(q^2 z; q) = -z. \quad (3.4)$$

Taking $z \rightarrow 1$ is now legitimate and gives

$$L = 2C_1 - C_2 - 1, \quad \mathcal{A} = L - C_1 + 2. \quad (3.5)$$

The pole term must not be dropped in this limit.

3.2 Solving the auxiliary generating function

Define $G(z) = \sum_{n \geq 0} Q_n z^n / (q; q)_n$. Directly from (2.3),

$$(1 - z)^2 G(z) = (1 + qz^2)G(qz), \quad G(0) = 1.$$

Iteration gives the exact identity

$$G(z) = \frac{(-qz^2; q^2)_\infty}{(z; q)_\infty^2}. \quad (3.6)$$

At $z = 1$, the coefficient of $(1 - z)^{-2}$ is $(-q; q^2)_\infty / (q; q)_\infty^2$. It follows, either by subtracting the principal part or by Lemma 3.1, that

$$\mathcal{L}_Q = \frac{(-q; q^2)_\infty}{(q; q)_\infty^3} = R(q). \quad (3.7)$$

The second expression for R in (1.6) follows from $(-q; q^2)_\infty = (q^2; q^2)_\infty^2 / ((q; q)_\infty (q^4; q^4)_\infty)$. In particular,

$$R(q) = \prod_{j \geq 1} (1 - q^j)^{-b_j}, \quad b_j = 4 - 2\mathbf{1}_{2|j} + \mathbf{1}_{4|j}. \quad (3.8)$$

This proves the colored-partition interpretation stated earlier.

4 A coefficientwise proof of the leading conjecture

4.1 An exact positive defect

Let $v_n = Q_n - 2h_n$. Then $v_0 = -1$, $v_1 = 0$, $v_2 = (1 - q)^2$. Its normalized pair at index 2 is

$$b_2 = e_2 = (1 - q^2)^{-2}.$$

The positive transfer lemma shows that its slope

$$E = R - 2L \quad (4.1)$$

has nonnegative coefficients. Combining (3.3) and (3.5) gives the exact decomposition

$$R - 4\mathcal{A} + 6 = E + 2C_2. \quad (4.2)$$

Both terms on the right are nonnegative power series. The constant 6 only corrects the chosen zero-area terms.

4.2 A positive majorant for the defect

We prove the bounds

$$0 \preccurlyeq E \preccurlyeq J, \quad 0 \preccurlyeq C_2 \preccurlyeq J + q + 2q^2. \quad (4.3)$$

These statements require a little care: a real-axis bound by $(1 - q)^2 R$ would not by itself imply a bound on coefficients.

First multiply the normalized Q solution by $(1 - q)^2$, and denote the resulting pair by $(\tilde{b}_n, \tilde{e}_n)$. At $n = 2$,

$$\begin{aligned} \tilde{b}_2 &= \frac{3 + 2q - q^2}{(1 - q^2)^2}, \\ \tilde{e}_2 &= \frac{1 + 2q + 3q^2 - 2q^4}{(1 - q^2)^2}. \end{aligned}$$

Subtracting the normalized pair for v gives

$$\frac{2 + 2q - q^2}{(1 - q^2)^2}, \quad \frac{2q + 3q^2 - 2q^4}{(1 - q^2)^2}.$$

Both are nonnegative power series: use $2 + 2q - q^2 = (1 + 2q) + (1 - q^2)$ and $3q^2 - 2q^4 = q^2 + 2q^2(1 - q^2)$. The transfer lemma and passage to slopes prove $E \preccurlyeq J$. The displayed pair for $\tilde{b}_2, \tilde{e}_2$ itself is nonnegative by the same decompositions.

Next, every coefficient of $(1 - q^n)h_n/(q; q)_n^2$ is nonnegative. For $n = 1$ this is immediate; for $n \geq 2$ it follows by removing only one of the two factors $(1 - q^n)$ in the denominator in (2.4). Hence $C_2 - C_3 \geq 0$. Setting $z = q$ in (3.4) gives

$$2C_2 - C_3 = (1 - q)^2 C_1 + q, \quad C_2 \preccurlyeq (1 - q)^2 C_1 + q.$$

It remains to compare the last expression with J .

Let $C_j^Q = \sum_{n \geq 0} Q_n q^{jn} / (q; q)_n^2$. For the Q solution the initial-condition term in (3.4) is zero, so $R = 2C_1^Q - C_2^Q$. The solution $u_n = Q_n - h_n$ has $u_0 = 0, u_1 = 1$; after normalization and multiplication by $(1 - q)^2$ its initial pair at 1 is $(1, 1)$. Consequently,

$$(1 - q)^2 (C_1^Q - C_1) \geq 0.$$

Also

$$(1 - q)^2 (C_1^Q - C_2^Q) = \sum_{n \geq 1} q^n (1 - q^n) \tilde{b}_n \geq -2q^2.$$

For completeness, the $n = 1$ summand is $2q - 2q^2$. For $n = 2$ the remaining factor is $(3 + 2q - q^2)/(1 - q^2) = 1 + (2 + 2q)/(1 - q^2)$, which is nonnegative. For $n \geq 3$, positivity follows from the nonnegative pair at index 2 and (2.4). These two inequalities imply $(1 - q)^2 C_1 \preccurlyeq J + 2q^2$, establishing (4.3).

Taking coefficients of (4.2) now proves (1.7). In particular, the finite bound is independent of any asymptotic approximation.

4.3 Why the squeeze is asymptotically sharp enough

The analytic estimates in Section 6 imply

$$p_n = 4c n^{-3/2} e^{B\sqrt{n}} \left(1 - \frac{d}{\sqrt{n}} + O(n^{-1}) \right), \quad (4.4)$$

$$j_n \sim \frac{K}{n} p_n. \quad (4.5)$$

Therefore the squeeze alone gives

$$a_n = c n^{-3/2} e^{B\sqrt{n}} \left(1 - \frac{d}{\sqrt{n}} + O(n^{-1}) \right).$$

This already proves the full proposed leading constant and the first correction. The further work below is needed for e and f , not for the leading conjecture.

5 Finite cuts: an exact rational approximation

5.1 General positive remainder series

The discrete Wronskian of h and Q is

$$h_m Q_{m+1} - h_{m+1} Q_m = (q; q)_m^2. \quad (5.1)$$

At $m = 0$ both sides are 1; the recurrence multiplies each succeeding Wronskian by $(1 - q^m)^2$, proving the identity by induction. Define

$$E_m = h_m R - Q_m L \quad (m \geq 1). \quad (5.2)$$

Thus $E_1 = E$.

Lemma 5.1 (A positive tail with a small radial bound). *The series E_m has nonnegative coefficients. For real $0 < q < 1$,*

$$0 \leq E_m(q) \leq (q; q)_m^2 R(q). \quad (5.3)$$

Furthermore, for each fixed m ,

$$C_{m+1}(e^{-t}) = O_m(t^{2m} R(e^{-t}) + 1) \quad (t \downarrow 0). \quad (5.4)$$

Proof. For $n \geq m$, the solution $w_n = h_m Q_n - Q_m h_n$ has $w_m = 0$, $w_{m+1} = (q; q)_m^2$. Its normalized pair at $m+1$ is $(1 - q^{m+1})^{-2}$ in both coordinates. Hence it is positive and its limiting slope, E_m , is positive.

For the real bound, let $z_m = 0, z_{m+1} = 1$ satisfy the original recurrence for later indices. Then $w_n = (q; q)_m^2 z_n$. For real q , the increments of Q_n are positive and nondecreasing:

$$Q_{n+1} - Q_n = (Q_n - Q_{n-1}) + (2q^n - q^{2n}) Q_{n-1}.$$

Thus $Q_n \geq n+1$ and $\Delta Q_m = Q_m - Q_{m-1} \geq 1$. A second Wronskian calculation yields

$$\frac{s_z}{s_Q} = Q_m \sum_{k=m}^{\infty} \frac{\prod_{j=m+1}^k (1 - q^j)^2}{Q_k Q_{k+1}} \leq \frac{1}{\Delta Q_m} \leq 1.$$

To obtain the middle inequality, use $Q_k \geq Q_m + (k - m)\Delta Q_m$ and telescope the reciprocals of the resulting arithmetic progression. Since normalization divides both slopes by $(q; q)_\infty^2$, (5.3) follows.

Finally, positivity gives $C_1 \leq C_1^Q \leq R$ on the real axis. Indeed, $Q - h$ is positive after normalization, and $R - C_1^Q = C_1^Q - C_2^Q \geq 0$. From (3.4) and $C_{j+1} \geq C_{j+2}$ on the real axis,

$$C_{j+1} \leq (1 - q^j)^2 C_j + q^j.$$

Iterate this inequality a fixed number m of times. The leading coefficient is $(q; q)_m^2 = O_m(t^{2m})$, and the finitely many remaining terms are bounded as $t \downarrow 0$. $\square$

5.2 Eliminating the auxiliary evaluations

Put $U_m = Q_m - h_m$, and let polynomials P_m be specified by

$$P_0 = P_1 = 0, \quad P_{j+1} = 2P_j - (1 - q^j)^2 P_{j-1} + q^j.$$

Iteration of (3.4) gives

$$C_{m+1} = Q_m C_1 - U_m(L + 1) - P_m.$$

Solving this identity together with (5.2) and $\mathcal{A} = L - C_1 + 2$ proves the following exact formula.

Proposition 5.2 (Finite-cut identity). *Where $Q_m(q) \neq 0$,*

$$\mathcal{A}(q) = \left(\frac{h_m}{Q_m} \right)^2 R - \frac{h_m E_m}{Q_m^2} - \frac{C_{m+1}}{Q_m} + 2 - \frac{U_m + P_m}{Q_m}. \quad (5.5)$$

After multiplication by Q_m^2 , this is an identity of analytic functions on the entire open unit disk and of formal power series.

The square in the first term is the precise source of the factor $1/4$: for every fixed $m \geq 1$, $h_m(1) = 2^{m-1}$ and $Q_m(1) = 2^m$. The identity also provides an algebraic expansion algorithm and a route to rigorous coefficient bounds.

For the coefficient theorem we use only $m = 3$:

$$\begin{aligned} h_3 &= 1 + 4q - q^4, & Q_3 &= 4 + 4q + 2q^2 - 2q^4 = -2(q + 1)(q^3 - q^2 - 2), \\ U_3 + P_3 &= 3 + 2q + 3q^2 - q^4, & F_3(q) &= \left(\frac{1 + 4q - q^4}{4 + 4q + 2q^2 - 2q^4} \right)^2. \end{aligned} \quad (5.6)$$

The cubic $q^3 - q^2 - 2$ has no zero in $|q| \leq 1$. A zero there would satisfy $|q|^2|q - 1| = 2$; equality in the bound $|q|^2|q - 1| \leq 2$ forces $q = -1$, which does not solve the equation. Thus $F_3 R$ is analytic in $|q| < 1$, with only a possible additional boundary pole at $q = -1$.

6 Saddle estimates for the comparison product

6.1 The local modular expansion

The classical eta transformation, in its Euler-product form, is

$$(e^{-z}; e^{-z})_\infty = \sqrt{\frac{2\pi}{z}} \exp\left(-\frac{\pi^2}{6z} + \frac{z}{24}\right) (e^{-4\pi^2/z}; e^{-4\pi^2/z})_\infty, \quad \Re z > 0. \quad (6.1)$$

The square root is the branch positive for positive real z . This follows from the usual modular transformation of Dedekind's eta function; see [5], equations 27.14.12–27.14.14. Applying it at $z, 2z, 4z$ gives, uniformly in any fixed closed sector $|\arg z| < \pi/2$,

$$R(e^{-z}) = \left(\frac{z}{2\pi}\right)^{3/2} \exp\left(\frac{K}{z} - \frac{z}{6}\right) \left(1 + O(e^{-\kappa/|z|})\right) \quad (6.2)$$

for an appropriate sector-dependent $\kappa > 0$. In particular, there is no omitted algebraic power-series correction in (6.2) beyond $\exp(-z/6)$.

6.2 A minor-arc bound rather than a Tauberian shortcut

Let $r = e^{-t}$. The colored-product form (3.8) implies

$$\frac{|R(re^{i\theta})|}{R(r)} \leq \exp\left[-\frac{c_0}{t} \min\left\{1, \frac{\theta^2}{t^2}\right\}\right] \quad (-\pi \leq \theta \leq \pi) \quad (6.3)$$

for all sufficiently small t and some $c_0 > 0$. Here and below a constant such as c_0 is unrelated to c in (1.1).

To verify (6.3), restrict the product to integers $N \leq j \leq 2N$, with N comparable to $1/t$. Then r^j stays in a compact subinterval of $(0, 1)$, and

$$\log \frac{|1 - r^j e^{ij\theta}|}{1 - r^j} \geq c_1(1 - \cos(j\theta)).$$

In the product for $|R(re^{i\theta})|/R(r)$, every omitted factor is at most 1. The elementary trigonometric-sum estimate

$$\sum_{j=N}^{2N} (1 - \cos(j\theta)) \geq c_2 N \min\{1, N^2 \theta^2\}$$

finishes the proof. For $|\theta| \leq 1/N$ it follows from the quadratic bound for $1 - \cos x$; for an intermediate compact range of $N\theta$ it follows by comparison with its strictly positive integral; for larger $N\theta$ use the geometric-sum bound for $\sum e^{ij\theta}$. This argument also covers the neighborhood of $\theta = \pi$.

Consequently a rational factor with at most fixed-order poles on the unit circle does not create a competing dominant saddle. The product's exponential suppression outweighs such polynomial losses.

6.3 The transfer calculation

For a local expansion

$$H(e^{-z}) \sim \lambda z^{3/2} e^{K/z} \sum_{j \geq 0} s_j z^j,$$

with the preceding minor-arc control, Cauchy's integral is localized at $t = \sqrt{K/n}$ and angular width $t^{3/2}$. Termwise saddle integration gives

$$[q^n]H(q) \sim \frac{\lambda K}{2\sqrt{\pi}} n^{-3/2} e^{2\sqrt{Kn}} \sum_{\ell \geq 0} n^{-\ell/2} \sum_{j+k=\ell} s_j K^{j/2} \frac{\beta_k(j)}{(2\sqrt{K})^k}, \quad (6.4)$$

where

$$\beta_k(j) = \frac{(-1)^k}{k!8^k} \prod_{r=1}^k ((5+2j)^2 - (2r-1)^2), \quad \beta_0(j) = 1. \quad (6.5)$$

For clarity, (6.4) asserts a finite-order expansion whenever a corresponding finite-order local expansion is available.

Here is a direct way to check the constants. The model integral for $z^{3/2+j}e^{K/z}$ is the Bessel-type contour expression

$$\left(\frac{K}{n}\right)^{(5/2+j)/2} L_{-5/2-j}(2\sqrt{Kn}).$$

Its large positive-argument saddle expansion is $e^x/\sqrt{2\pi x}$ times the polynomial coefficients (6.5) in powers of x^{-1} . Alternatively those same coefficients follow by substituting a power series into the Bessel differential equation. The coefficient $\lambda K/(2\sqrt{\pi})$ in (6.4) follows immediately. One need not identify the original generating function with a Bessel function: only this local contour model is used.

For a fully controlled remainder, take a central arc $|\theta| \leq t^{3/2-\delta}$ with $0 < \delta < 1/6$. On the complement, (6.3) makes the integral smaller than any algebraic multiple of its leading saddle value. On the central arc, put $\theta = t^{3/2}u$; the real part of the quadratic term is a negative constant times u^2 , and Taylor remainders are bounded by a polynomial in u times that Gaussian. Truncation and integration are therefore valid to every fixed order. This supplies the error justification behind (6.4).

For $H = R$, use $\lambda = (2\pi)^{-3/2}$ and $s_j = (-1/6)^j/j!$. This proves (4.4). Multiplication by $(1 - e^{-z})^2 = z^2 + O(z^3)$ gives (4.5). It also gives

$$p_n = 4c n^{-3/2} e^{B\sqrt{n}} \left(1 - \frac{d}{\sqrt{n}} + \frac{\frac{1}{2} + \frac{3}{4K} + \frac{K}{72}}{n} + O(n^{-3/2}) \right). \quad (6.6)$$

The comparison product itself has an all-orders coefficient expansion; this fact is not being inferred for $\mathcal{A}$ without additional estimates.

7 Proof of the three correction terms

7.1 Bounding the finite-cut remainder on the whole circle

Proposition 7.1. *With F_3 from (5.6),*

$$a_n = [q^n](F_3 R) + O(p_n n^{-7/4}). \quad (7.1)$$

Proof. Take $q = re^{i\theta}$ with $r = e^{-t}$ and $t = \sqrt{K/n}$. By positivity and Lemma 5.1,

$$|E_3(q)| \leq E_3(r) = O(t^6 R(r)), \quad |C_4(q)| \leq C_4(r) = O(t^6 R(r) + 1).$$

Since the cubic factor of Q_3 is bounded away from zero on the closed unit disk, there are uniform estimates

$$\int_{-\pi}^{\pi} \frac{d\theta}{|Q_3(re^{i\theta})|^2} = O(t^{-1}),$$

$$\int_{-\pi}^{\pi} \frac{d\theta}{|Q_3(re^{i\theta})|} = O(\log(1/t)).$$

The first follows from $\int_{-\pi}^{\pi} |1 + re^{i\theta}|^{-2} d\theta = 2\pi/(1 - r^2)$; the second follows by integrating $(t^2 + (\theta - \pi)^2)^{-1/2}$ near the single boundary zero. The polynomials h_3, U_3, P_3 are uniformly bounded.

Apply Cauchy's coefficient formula to the three remainder terms in (5.5) with $m = 3$. Their total is

$$O(r^{-n}\{t^5 R(r) + t^6 R(r) \log(1/t) + \log(1/t) + 1\}).$$

The terms without $R(r)$ are exponentially smaller than the main coefficient scale. Since the saddle estimate gives $p_n \asymp r^{-n} R(r) t^{3/2}$, the dominant error is $O(p_n t^{7/2}) = O(p_n n^{-7/4})$. $\square$

7.2 Extracting the coefficients

A finite rational expansion at $q = 1$ gives

$$4F_3(e^{-z}) = 1 - \frac{1}{2}z^2 + \frac{1}{2}z^3 + O(z^4).$$

Together with (6.2), this yields

$$(F_3 R)(e^{-z}) = \frac{z^{3/2} e^{K/z}}{4(2\pi)^{3/2}} \left(1 - \frac{z}{6} - \frac{35z^2}{72} + \frac{755z^3}{1296} + O(z^4) \right), \quad (7.2)$$

up to a locally exponentially small modular remainder. The rational factor has only fixed-order boundary poles, so the minor-arc argument applies. Using (6.4) through $\ell = 3$ gives (1.3)–(1.5); its own truncation error is $O(n^{-2})$ relative to the leading term. The larger remainder in Proposition 7.1 is $O(n^{-7/4})$, proving Theorem 1.1.

For an explicit audit of the algebra, the relative n^{-1} coefficient is

$$\underbrace{\frac{3}{4K}}_{j=0, k=2} + \underbrace{\frac{1}{2}}_{j=1, k=1} - \underbrace{\frac{35K}{72}}_{j=2, k=0}.$$

The relative $n^{-3/2}$ coefficient is

$$\underbrace{-\frac{5}{8\sqrt{K}}}_{j=1, k=2} + \underbrace{\frac{175\sqrt{K}}{72}}_{j=2, k=1} + \underbrace{\frac{755K^{3/2}}{1296}}_{j=3, k=0}.$$

There is no $j = 0, k = 3$ contribution: the Bessel polynomial for that half-integer order terminates after $k = 2$.

Subtracting four times (1.2) from (6.6) proves (1.8). Thus the lost proportion in the colored-partition comparison is asymptotically $K/(2n)$, substantially smaller than the explicit upper bound $3K/n$ supplied by the elementary squeeze.

8 A proved all-orders radial expansion

8.1 A compatible family of rational approximants

Define $D(q) = L(q)/R(q)$ for real $0 < q < 1$. For every fixed $m \geq 1$, (5.2) and (5.3) imply

$$D(e^{-t}) = \frac{h_m(e^{-t})}{Q_m(e^{-t})} + O_m(t^{2m}). \quad (8.1)$$

The denominator tends to 2^m , so it is nonzero near the positive real limit. The Taylor expansions of these rational functions are therefore compatible: the coefficient of t^j is fixed once $2m > j$. Let $\mathcal{D}(t)$ be their unique formal limit.

Theorem 8.1 (Radial expansion to arbitrary fixed order). *The expansion (1.9) holds with*

$$\mathcal{F}(t) = 4\mathcal{D}(t)^2, \quad \mathcal{D}(t) = \lim_{m \rightarrow \infty}^{\text{formal}} \frac{h_m(e^{-t})}{Q_m(e^{-t})}. \quad (8.2)$$

For each fixed m ,

$$\frac{\mathcal{A}(e^{-t})}{R(e^{-t})} = \left(\frac{h_m(e^{-t})}{Q_m(e^{-t})} \right)^2 + O_m(t^{2m}). \quad (8.3)$$

The coefficients through t^8 are those in (1.10).

Proof. Divide the finite-cut identity (5.5) by R . For fixed m , the denominator $Q_m(e^{-t})$ stays bounded away from zero. The E_m/R and C_{m+1}/R terms are $O_m(t^{2m})$ by Lemma 5.1; the remaining rational expression divided by R is smaller than every power of t by (6.2). This proves (8.3). Compatibility and the product expansion give (8.2). Ordinary rational series arithmetic then gives the displayed coefficients. $\square$

For example, setting $w = 1 - q$ gives the useful intermediate expression

$$\mathcal{D} = \frac{1}{2} - \frac{1}{8}w^2 - \frac{5}{32}w^4 + \frac{1}{8}w^5 - \frac{65}{128}w^6 + \frac{37}{32}w^7 - \frac{2053}{512}w^8 + O(w^9).$$

Replacing w by $1 - e^{-t}$ and squaring reproduces (1.10). The accompanying symbolic checker verifies these operations exactly over the rationals.

8.2 Why this is not yet an all-orders coefficient theorem

Increasing m makes the positive radial tails smaller, but the complex behavior of $1/Q_m$ must still be controlled. For general m , one cannot simply assume that all zeros of Q_m lie outside the unit disk. No general zero classification is assumed or established here. A meromorphic rational approximant can be useful on a contour, but poles inside that contour require attention if one interprets its integral as an ordinary Taylor coefficient.

The choice $m = 3$ deliberately avoids these issues: the factorization in (5.6) proves the required zero exclusion exactly. Consequently the first three correction terms are supported by a full complex argument. Coefficients at later orders obtained by purely formal transfer of (1.9) are candidates until an equally valid remainder argument is supplied. No exponentially small contribution from other roots of unity, nor a complete transseries with all such sectors, is claimed in this report.

[write] Added 2 October 2026, batch 77P5: the gap is closed in Part II. Part II proves the all-orders coefficient theorem (Theorem 14.1): for every fixed M , a_n has an expansion in powers of $(n - 1/6)^{-1/2}$ with remainder $O(n^{-M/2})$ and algorithmically determined coefficients. It does not control the zeros of Q_m . It avoids them: the series $D = L/R$ of this section is a partial theta function (Remark 16.2), and $\mathcal{A} = R D^2$ plus a remainder whose coefficients are exponentially smaller (Proposition 16.1, Corollary 17.2), with whole-circle bounds for both factors. The later coefficients called “candidates” above are thereby proved. The statement about exponentially small sectors and a complete transseries still stands for both parts.

9 Inverse asymptotics and integer thresholds

9.1 A real inverse requires an interpolation convention

The sequence itself is defined only on integers. To state a real inverse accurately, choose an interpolation that preserves the proved asymptotic accuracy. Put

$$M(x) = c x^{-3/2} e^{B\sqrt{x}} (1 - dx^{-1/2} + ex^{-1} + fx^{-3/2})$$

for sufficiently large real x , where the parenthesis is positive. For an integer k , set $r_k = \log a_k - \log M(k)$. Theorem 1.1 gives $r_k = O(k^{-7/4})$. For $x = k + \theta$, $0 \leq \theta \leq 1$, define

$$\log \tilde{a}(x) = \log M(x) + (1 - \theta)r_k + \theta r_{k+1}. \quad (9.1)$$

This continuous interpolation agrees with a_k at integers and obeys

$$\log \tilde{a}(x) = \log M(x) + O(x^{-7/4}).$$

It is eventually strictly increasing: the derivative of $\log M$ is $B/(2\sqrt{x}) + O(x^{-1})$, while the piecewise-linear residual has slope $O(x^{-7/4})$. Denote its eventual inverse by $v(y)$. Other interpolations with the same asymptotic error give the same expansion below to the stated accuracy.

This convention matters beyond low orders. Direct linear interpolation of $\log a_k$, without first subtracting $\log M$, introduces a curvature error of order $x^{-3/2}$, which can obscure the third logarithmic correction. Equation (9.1) avoids that ambiguity.

9.2 Lambert W and the correction terms

Define

$$\begin{aligned} D_0 &= Bd = \frac{B^2}{12} + 3, \\ \alpha_2 &= B^2(e - d^2/2), \quad \alpha_3 = B^3(f + de - d^3/3), \\ \ell &= \log \frac{y}{cB^3}, \quad s_0 = -3W_{-1}\left(-\frac{1}{3}e^{-\ell/3}\right). \end{aligned} \quad (9.2)$$

Here W_{-1} is the real lower branch, which tends to $-\infty$ as its argument tends to 0 from below. It is selected because the inverse index tends to $+\infty$.

Theorem 9.1 (Asymptotic inverse). *For the interpolation (9.1), as $y \rightarrow \infty$,*

$$v(y) = \frac{1}{B^2} \left[s_0^2 + 2D_0 + \frac{2(3D_0 - \alpha_2)}{s_0} + \frac{18D_0 - 6\alpha_2 - D_0^2 - 2\alpha_3}{s_0^2} \right] + O(s_0^{-5/2}). \quad (9.3)$$

In particular, a formula involving only logarithms is

$$v(y) = \frac{\ell^2 + 6\ell \log \ell + 9(\log \ell)^2 + 18 \log \ell + 2D_0}{B^2} + O\left(\frac{(\log \ell)^2}{\ell}\right). \quad (9.4)$$

Proof. Set $s = B\sqrt{v(y)}$. Taking logarithms in the coefficient expansion and using the interpolation error gives

$$\ell = s - 3 \log s - \frac{D_0}{s} + \frac{\alpha_2}{s^2} + \frac{\alpha_3}{s^3} + O(s^{-7/2}). \quad (9.5)$$

The definition of s_0 is equivalent to $s_0 - 3 \log s_0 = \ell$. Put $s = s_0 + \delta$ and compare successive inverse powers of s_0 . Since $1 - 3/s_0$ is bounded away from zero,

$$\delta = \frac{D_0}{s_0} + \frac{3D_0 - \alpha_2}{s_0^2} + \frac{9D_0 - 3\alpha_2 - D_0^2 - \alpha_3}{s_0^3} + O(s_0^{-7/2}).$$

Squaring and dividing by B^2 gives (9.3). For (9.4), direct substitution in (9.5) yields

$$s = \ell + 3 \log \ell + \frac{9 \log \ell + D_0}{\ell} + O((\log \ell)^2 / \ell^2),$$

and another squaring completes the calculation. □

If $N(y) = \min\{n \geq n_0 : a_n \geq y\}$ for a sufficiently large fixed n_0 , then exactly

$$N(y) = \lceil v(y) \rceil.$$

The error term belongs *inside* that ceiling. Uniformly discarding it can produce an incorrect integer when the approximation is very close to a threshold. A safe coarser statement is that the right-hand side of (9.4), without its remainder, differs from $N(y)$ by $O(1)$. Away from integers by more than a proved error bound, rounding would be unambiguous; this article does not supply a numerical universal constant for that error bound.

[write] Added 2 October 2026, batch 75: an instance of repository results. The inversion of this section is an instance of the Lambert-centred inversion apparatus of the transseries-and-inversion volume [8], and no novelty is claimed for the method. The core $s_0 - 3 \log s_0 = \ell$ is `p0:thm:lambert-core` with $a = 1$, $b = -3$; its branch rule selects W_{-1} , as above. Equation (9.5) is the core plus a power series in $1/s$, and the correction δ is the all-orders reversion around the core, `p0:thm:lambert-centered`, truncated at the order the forward remainder allows. The shape is that of the partition-number chapter of the same volume, `p3:sec:lambert` with `p3:thm:core-correction`, where the factor n^{-1} in $p(n)$ gives $b = -2$; here $n^{-3/2}$ gives $b = -3$. The rule $N(y) = \lceil v(y) \rceil$ and the warning that follows it are part (1) and part (2) (the separation condition) of `p0:thm:staircase`, specialized as in `p3:thm:staircase`. The generic arithmetic of those two parts is formalized in Lean as `Fabius.staircase_ceil` and `Fabius.staircase_separation` in `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`; those statements concern an arbitrary monotone interpolation and give no formal status to Theorem 9.1. What is specific to this report is the forward expansion that feeds the inversion and the explicit interpolation (9.1).

10 Reproducible exact and numerical checks

10.1 What was computed

The attached `verify.py` uses only Python's standard library and unbounded integers. It computes $a_0, \dots, a_{2000}$ from the normalized recurrence, and independently computes p_n from the colored Euler product. Division by $(1 - q^m)^2$ is performed by two exact cumulative-sum passes. Thus the computation avoids floating-point cancellation in the original polynomial recurrence.

The checker confirms the initial 37 OEIS terms and four separately read b-file anchors, at $n = 50, 100, 200, 500$ [4]. It then verifies the slope identity, the positive defect decomposition, both majorants in (4.3), and the squeeze for every integer $3 \leq n \leq 2000$. The third finite-cut identity, after clearing its polynomial denominator, and positivity of its auxiliary tails are checked as well. The output CSV contains all computed coefficients, not just a selected plot. This is an exact finite test, not a formal proof of the infinite statements.

The optional `verify_symbolic.py` uses SymPy for rational symbolic arithmetic and mpmath for high-precision numerical evaluations. It checks Wronskians, the factorization of Q_3 , the radial coefficients through t^8 , and the transfer coefficients d, e, f . All included assertions passed in the supplied run.

10.2 Accuracy of the asymptotic approximants

Let

$$A_0(n) = cn^{-3/2}e^{B\sqrt{n}}, \quad A_1 = A_0(1 - d/\sqrt{n}), \quad A_2 = A_1 + A_0e/n, \quad A_3 = A_2 + A_0f/n^{3/2}.$$

The table reports $a_n/A_j(n)$; a ratio approaching 1 is the desired behavior. Rounded display values are not used by the checker.

n	a_n/A_0	a_n/A_1	a_n/A_2	a_n/A_3
50	0.836729727	0.980057860	1.027184142	0.984290725
100	0.885790139	0.987955090	1.010017535	0.995767525
200	0.920485946	0.993103974	1.003708062	0.998875075
500	0.950776376	0.996878614	1.000989560	0.999807957
1000	0.965688969	0.998335826	1.000361244	0.999950135
2000	0.976025253	0.999128395	1.000130943	0.999987161

More terms need not improve a short truncation at a modest index: $n = 50$ illustrates this. At $n = 2000$, the four-term relative error is approximately 1.284×10^{-5} . The data also show

$$n(1 - 4a_n/p_n) = 1.739699, 1.940375, 2.110009, 2.286402, 2.386995, 2.463821$$

at the six listed indices, tending toward the proved limit $K/2 = 2.673017858628 \dots$

10.3 Testing the inverse at known sequence values

For $y = a_n$, the compatible inverse is exactly n . The leading Lambert approximation is s_0^2/B^2 ; the corrected one is the bracketed expression in (9.3).

n	Leading inverse minus n	Corrected inverse minus n
50	-0.598588435	-0.028604645
100	-0.560169154	-0.011441486
200	-0.530798365	-0.004479985
500	-0.502617905	-0.001257271
1000	-0.487447241	-0.000471103
2000	-0.476247422	-0.000174032

The nonvanishing limiting offset in the uncorrected Lambert approximation is explained by the constant term $2D_0/B^2$ in the inverse theorem. These values are numerical checks of already derived formulas.

11 Further research questions and concrete next steps

1. **An all-orders coefficient expansion.** Use the general finite-cut identity (5.5) to extend the $O(n^{-7/4})$ remainder to arbitrary powers. The outstanding issue is not generating formal coefficients; it is uniform contour control of Q_m^{-1} and treatment of possible interior poles of rational approximants. A theorem organizing admissible cuts would turn the radial algorithm into an all-orders coefficient algorithm.

[write] Added 2 October 2026, batch 77P5: answered. Part II proves the all-orders coefficient expansion (Theorem 14.1) by a different route, an exact product-square decomposition, without organizing finite cuts. A theorem on admissible cuts, or on the zeros of Q_m , remains a separate question.

2. **Explicit numerical error constants.** Make the product minor-arc constants and the cubic lower bound in the $m = 3$ proof effective. This would give certified numerical intervals for a_n and safe integer inversion, rather than asymptotic O statements with unspecified constants.
3. **A combinatorial explanation of the factor four.** Construct a direct relation between four copies of the L-convex class and the colored partitions counted by R . The proven inequality $p_n \geq 4a_n$ for $n \geq 1$ is suggestive, but an injection or decomposition is not supplied by the analytic proof. A description of the defect class could explain the limit $K/2$ combinatorially.
4. **The singularities of the boundary series.** Study $D(q) = L/R$ as an analytic object, not just a radial formal series. The recurrence, Wronskians, and positive tails give several exact representations. Determine its domain of continuation and the role of zeros of the finite denominators Q_m .

[write] Added 2 October 2026, batch 77P5: partly answered. $D = L/R$ equals the partial theta function $1 - \sum_{k \geq 0} q^{k(k+1)}(1 - q^{2k+1})/(1 + q^{2k+1})$ on the whole disk $|q| < 1$ (Remark 16.2, from Part II's Proposition 16.1). Its continuation and the zeros of Q_m remain open; Part II's open problem 5 continues the question.

5. **Beyond-all-orders terms.** Separate exponentially small contributions to $\mathcal{A}(e^{-t})$ from the modular product and from the recurrence boundary corrections. Determine which roots of unity contribute to a coefficient transseries and whether the radial series has a useful summability or resurgence structure. None of these properties follows just from the computed rational coefficients.

[write] Added 2 October 2026, batch 77P5. Part II separates an exact remainder whose coefficients are $O(n^{1/4}e^{\pi\sqrt{2}n})$, exponentially smaller than a_n (Corollary 17.2); the bound is deliberately coarse. The actual exponentially small contributions remain open (Part II, open problem 2).

6. **A reusable theorem for recurrence-defined area series.** Abstract the positive-transfer and finite-cut mechanism to other recurrences $y_{n+1} = a(q^n)y_n - b(q^n)y_{n-1}$ whose auxiliary generating functions are Euler products. State conditions under which a positive boundary defect is small enough for coefficient transfer. The present proof gives a worked example with all analytic and positivity obligations visible.
7. **Z-convex and higher-convexity area enumeration.** Investigate whether the numerically studied Z-convex area asymptotics [6] admit a related positive comparison. This is a proposed application,

not an assertion that the L-convex recurrence or constant extends unchanged. Area and perimeter must remain distinct parameters.

[write] **Added 2 October 2026, batch 77P5.** Part II's open problem 4 asks the analogous question for its product-square decomposition.

8. **Formalization and OEIS integration.** Formalize the normalized transfer, coefficientwise order, Wronskian, and finite-cut identity first. These are algebraic and should be separated from the eta transformation and complex saddle analysis. An OEIS update should cite a reviewed version of the coefficient proof, rather than only a numerical output or the all-orders radial statement.

12 Conclusion and claim ledger

The principal result is a proof of a specific recorded conjecture, not a conjecture inferred from additional terms. The established generating function leads to a positive normalized recurrence; one auxiliary initial condition leads exactly to a colored-partition Euler product. Two boundary conditions produce the factor $1/4$. A positive coefficient squeeze proves the leading asymptotic and its first correction. A finite cut at index three, together with a full-circle error estimate, proves two further corrections and a relative $O(n^{-7/4})$ remainder.

The following boundaries summarize what can be reused without confusion.

Statement	Status in this manuscript
Area generating function and its enumerative interpretation	Cited input from prior work
Colored-partition product, positive squeeze, finite-cut identity	Proved here from the recurrence
Leading coefficient asymptotic and three correction terms	Proved with a complex remainder estimate
All-orders expansion for $\mathcal{A}(e^{-t})$, $t > 0$	Proved to every fixed radial order
Inverse expansion for a specified compatible interpolation	Proved; integer rounding treated separately
All-orders coefficient expansion and exponentially small sectors	Further research; not claimed
Finite checks through $n = 2000$ and symbolic arithmetic	Executed checks, not formal certification
Independent peer review or Lean verification	Not performed in this report

[write] **Added 2 October 2026, batch 77P5.** The ledger is Part I's. Its row "All-orders coefficient expansion and exponentially small sectors" is now half answered: Part II proves the all-orders coefficient expansion (Theorem 14.1); exponentially small sectors are still not claimed. The last two rows hold for Part II as well.

Part II. All-order asymptotics and inversion

Sections 1–12 and Appendices A and B form the original report (Part I) of 1 October 2026. Part I is unchanged except for dated [write] pointers to this addition (after the scope box, in Section 1.1, at the end of Section 8.2, in research questions 1, 4, 5 and 7 of Section 11, and after the claim ledger of Section 12), a line in the title, contents entries for the two parts, two macros in the preamble, one bibliography entry used only by Part II, and notes in two existing entries. No number of Part I moves.

13 Part II: source, scope and notation

This section was written when Part II was added. Sections 14–22 print the manuscript; Section 23 records the provenance of both parts and the choices the merge made.

13.1 Source and status

Sections 14–22 print, in full, manuscript 31 of batch 77 of ProveIt’s incoming-reports intake, titled “All order asymptotics and inversion for L convex polyominoes” and dated 1 October 2026. It arrived as the archive `lconvex-asymptotics-reproducibility.zip` in commit `096ee7b87` and was placed in commit `4f11bc9c0`. Its author line is empty and it names no tool; like every manuscript of the intake it is treated as AI-assisted research output. It names no ProveIt commit and does not mention the repository, so it has no pin, and it was not written against Part I.

It is not a version of Part I’s manuscript. Only 40 of its 380 distinct non-blank source lines occur in Part I’s source. Its proof architecture is different: an exact product-square decomposition and a Hankel–Bessel transfer, where Part I uses a positive coefficient squeeze and finite cuts. The proof note that its shipped audit reviewed is a different file from Part I’s manuscript. The package ships three internal reviews (`02-all-orders-mathematical-audit.md`, `02-all-orders-final-source-review.md`, `02-all-orders-computational-review.md`). They are the delivery’s own checks, not external peer review. Part II is unrefereed. Nothing in it is formalized in Lean or Rocq, and its place in the collection confers no formal status on it.

What was rechecked at intake and in this write. Independently of the delivery:

- $a_0, \dots, a_{120}$ from the original form (14.1) agree with the shipped coefficient file, and that file agrees entry by entry with the column a_n of Part I’s `data/coefficients.csv` for all $0 \leq n \leq 2000$;
- $c_0, \dots, c_4$ follow from (14.6) with $b_0, \dots, b_4$ read from (18.1), the h_r of Section 20 are $(2\sqrt{\kappa})^r c_r$, and the unshifted re-expansion gives exactly Part I’s d, e, f (Remark 14.2), all with exact SymPy arithmetic;
- $B = D$ and $A = PB^2 + R$, with R from the literal $E(q)$, at $q = 0.3, 0.6$ and $0.3 + 0.4i$, to 40 digits (discrepancies 10^{-38} to 10^{-41});
- the identity of Remark 16.2 between Part I’s L/R^1 and Part II’s B , at the same three points;
- the entries of Table 2, the leading constant, the monotonicity injection (by hand), and the log-concavity expansion numerically (Corollary 19.1);
- the shipped replay, core and optional parts (Section 21).

The whole-circle estimates (Lemmas 17.1 and 17.3) and the contour deformation in Section 19 were read but not re-derived in detail; they are written at the level of standard saddle-point estimates.

13.2 What Part II adds, and what stays in Part I

Part II proves, from the same cited area generating function:

- the exact decomposition $A = PBD + R$ and the identity $B = D$, a partial theta function, hence $A = PB^2 + R$ exactly (Propositions 15.2 and 16.1);
- whole-circle bounds and an exponentially smaller exact remainder (Lemma 17.1, Corollary 17.2);
- **the all-orders coefficient theorem** (Theorem 14.1) in the shifted index $N = n - 1/6$, with coefficients given by a finite algorithm. This closes the gap that Part I states in Section 8.2 and research question 1 of Section 11;
- eventual strict log-concavity with a two-term expansion of $\log(a_n^2/(a_{n-1}a_{n+1}))$ (Corollary 19.1);
- an injection proving that a_n is nondecreasing;
- a Lambert-W inverse in shifted coordinates, logarithmic inverse polynomials, and shrinking integer brackets for every order (Section 20).

Only Part I has the coefficientwise squeeze $0 \leq p_n - 4a_n \leq 3j_n$ for every $n \geq 3$ with its colored-partition reading, the deficit limit $K/2$, the finite-cut identity and its $m = 3$ route, the interpolation $\nu(y)$ with the exact rule $N(y) = \lceil \nu(y) \rceil$, and the tables of Section 10.

Several results are proved in both parts, by different arguments; each part keeps its own proof.

- The leading asymptotic and the corrections d, e, f of Theorem 1.1: Part I by squeeze and finite cut, with remainder $O(n^{-7/4})$; Part II as the case $M = 4$ of Theorem 14.1, with remainder $O(n^{-2})$ (Remark 14.2).
- The radial expansion (1.9) with the coefficients (1.10) of Theorem 8.1: it follows from (16.2), and the coefficients are four times those of $D(e^{-z})^2$ in (18.1) (Remark 16.2).
- The local eta expansion (6.2) is (18.3); the minor-arc bound (6.3) is Lemma 17.3, for the same product; the transfer (6.4)–(6.5) is (19.2)–(14.6) in the shifted index.
- The generating function (2.1) is (14.1), with Part I's h_n equal to the manuscript's f_{n-1} ; the initial terms are printed in both parts.

13.3 Notation across the two parts

Part II keeps the manuscript's letters. The shipped data files and reviews use them (for example the keys `A_equals_P_B_squared_plus_R` and `initial_R` of `02-all-orders-exact-identities.json`, and `kappa` in the forward and inverse series), and renaming would break that correspondence. Many of these letters mean something else in Part I. In Sections 14–22 a letter has its Part II meaning. Where a [write] note in Part II refers to a Part I object whose letter clashes, it adds the superscript I: $R^I, K^I, B^I, D^I, L^I, y^I$. No symbol of either manuscript was renamed and no normalization was changed. Table 1 lists every clash. The last column gives the tempting false reading.

Table 1: Letters of Part II, the same object in Part I, and the clash to avoid.

Part II	Meaning in Part II	Same object in Part I	Not to be confused with
κ	$13\pi^2/24$	$K, (1.1)$	Part I's κ in (6.2), an unspecified sector-dependent exponent
C	$13\sqrt{2}/768$	c	Part I's series $C(z; q)$, C_j , (3.2)
N	$n - 1/6$	—	Part I's threshold $N(y)$, and its block length $N \asymp 1/t$ in the minor-arc proof
$P(q)$	$(-q; q^2)_\infty / (q; q)_\infty^3$, (15.3), (18.2)	$R(q)$, (1.6)	Part I's polynomials P_m ; Part II's P_j
$R(q)$	exact remainder $1 + E(q)/(q; q)_\infty$, Proposition 15.2	none	Part I's $R(q)$, which is Part II's P : $A = PB^2 + R$ does <i>not</i> say $\mathcal{A} = R^1 D^2 + R^1$
p_n	$[q^n]P(q)B(q)^2$, Corollary 17.2; $p_n \sim a_n$	none	Part I's $p_n = [q^n]R^1(q) = [q^n]P(q)$, with $p_n \sim 4a_n$
$E(q)$	the double sum (15.6)	none	Part I's defect $E = R - 2L$, (4.1), and tails E_m , (5.2)
$B(q)$	$1 - \sum_l u_l$, the partial theta function (16.1)	$D(q) = L/R$ of Section 8.1, exactly (Remark 16.2)	Part I's constant $B = 2\sqrt{K} = \pi\sqrt{13/6}$
$D(q)$	the series of (15.3); $D = B$	$D(q) = L/R$: the same function, not only the same letter	Part II's bracket constants D_M
K	$C(2\sqrt{\kappa})^3$, (20.1)	cB^3	Part I's $K = 13\pi^2/24$, which is Part II's κ
L, w	$\log(X/K)$, $\log L$	$\ell = \log(y/(cB^3))$ of (9.2)	Part I's slope series $L = \mathcal{L}_h$, (3.1)
X	target value of the inverse	y of Section 9	—
y	$2\sqrt{\kappa(n - 1/6)} = B^1\sqrt{N}$	unshifted analogue $s = B\sqrt{v}$	Part I's y , the target value
y_0	Lambert root (20.4)	s_0 of (9.2), the same number when $X = y^1$	—
$n_*(X)$	$\min\{n : a_n \geq X\}$	$N(y)$ of Section 9	—
F_M, x_M	real model (20.7) and its large inverse	analogues of $M(x)$ and $v(y)$	Part I's rational $F_3(q)$, (5.6), and series $\mathcal{F}(t)$
M	truncation order	—	Part I's model function $M(x)$
h_n, H, G	$f_{n-1}/(q; q)_n$, its generating function, the product (15.1)	Part I's $h_n = f_{n-1}$ lacks the factor $1/(q; q)_n$; G is Part I's G , (3.6)	h_r of Section 20
h_r, ℓ_r	coefficients of (20.2) and of its logarithm, (20.3)	D_0, α_2, α_3 play this role for s	h_n above; Part I's ℓ
b_m	Taylor coefficients of $D(e^{-z})^2$, (14.5)	$\mathcal{F}(t) = 4 \sum b_m t^m$, (8.2)	Part I's $b_n = y_n/(q; q)_n^2$ and color multiplicities b_j , (3.8)
c_r	coefficients of (14.4)	unshifted: $1, -d, e, f$	Part I's c and c_0 ; the local constants $c, c_1, \dots, c_4, C_1$ of Part II's proofs
u_l, T_j	summands of B in (15.3); summands of D (proof of Lemma 18.1)	—	Part I's $u_n = Q_n - h_n$ and $T_n(q)$

Part II	Meaning in Part II	Same object in Part I	Not to be confused with
P_j	logarithmic inverse polynomials	—	$P(q)$; Part I's P_m
a, b, S, V	local to the proof of Proposition 16.1 ($a = i\sqrt{q}$)	—	$a_n; b_m$; Part I's s
z, t	auxiliary variable in Section 15; from Section 18 on $q = e^{-z}$, $t = \Re z$	same two uses in Part I	saddle radius $t = \sqrt{\kappa/N}$ here, $t = \sqrt{K/n}$ in Part I
λ, α	$\sqrt{\kappa N}; 3/2 + m$	—	Part I's amplitude λ in (6.4); α_2, α_3
d_r	local to the proof of Corollary 19.1	—	Part I's d

Two further letters are shared with the same meaning: f_n (the recurrence (14.2); Part I's h_{n+1}) and the branch W_{-1} .

13.4 The manuscript's abstract

Let a_n count fixed L -convex polyominoes of area n , with the empty object included. Starting from the known exact area generating function, we prove the asymptotic equivalent conjectured by Guttmann and Kotěšovec and recorded in OEIS A126764. An exact decomposition into an eta quotient times the square of a partial theta function, plus an exponentially smaller coefficient remainder, supplies every fixed algebraic order. The correction coefficients are given by a finite algorithm. We also give a branch-explicit Lambert W inverse, logarithmic inverse polynomials, and shrinking intervals for the integer threshold inverse before rounding. The proof includes whole-circle estimates; a positive-real singular expansion alone is not used as a coefficient-transfer argument. Exact coefficient checks and a portable replay accompany the analysis.

14 The sequence and the main theorem

An L -convex polyomino is a convex polyomino in which any two cells can be joined internally by a path with at most one right-angle turn. We count fixed polyominoes, identifying translations but keeping rotations and reflections distinct. Its area generating function is

$$A(q) = \sum_{n \geq 0} a_n q^n = 1 + \sum_{n \geq 1} \frac{q^n f_{n-1}(q)}{(q; q)_{n-1}^2 (1 - q^n)}, \quad (14.1)$$

where

$$f_{-1} = f_0 = 1, \quad f_n = 2f_{n-1} - (1 - q^n)^2 f_{n-2} \quad (n \geq 1). \quad (14.2)$$

Here $(a; q)_j = \prod_{h=0}^{j-1} (1 - aq^h)$. Formula (14.1) is the Castiglione–Frosini–Munarini–Restivo–Rinaldi area formula [2], in the normalization reproduced in [3, Eq. (1.1)]. The first terms are

$$1, 1, 2, 6, 15, 35, 76, 156, 310, 590, 1098, 1984, 3515, 6094, \dots$$

Set

$$\kappa = \frac{13\pi^2}{24}, \quad C = \frac{13\sqrt{2}}{768}, \quad N = n - \frac{1}{6}. \quad (14.3)$$

Theorem 14.1. *For every fixed integer $M \geq 1$,*

$$a_n = CN^{-3/2}e^{2\sqrt{\kappa}N} \left(\sum_{r=0}^{M-1} c_r N^{-r/2} + O(N^{-M/2}) \right). \quad (14.4)$$

The coefficients are algorithmically determined as follows. Define the formal asymptotic coefficients b_m by

$$\left(\sum_{j \geq 0} \frac{(-1)^j e^{-zj(j-1)/2} (e^{-z}; e^{-z})_j}{(-e^{-z}; e^{-2z})_{j+1}} \right)^2 \sim \sum_{m \geq 0} b_m z^m. \quad (14.5)$$

To determine $b_0, \dots, b_{M-1}$, it suffices to retain $0 \leq j < M$ on the left and take ordinary Taylor coefficients. Then

$$c_r = \sum_{\substack{m+k=r \\ 0 \leq k \leq m+2}} 4b_m \kappa^{m/2} \frac{(-1)^k (m+2+k)!}{(m+2-k)! k! (4\sqrt{\kappa})^k}. \quad (14.6)$$

In particular,

$$\begin{aligned} c_0 &= 1, & c_1 &= -\frac{3}{2\sqrt{\kappa}}, & c_2 &= \frac{3}{4\kappa} - \frac{\kappa}{2}, \\ c_3 &= \frac{\sqrt{\kappa}(\kappa+5)}{2}, & c_4 &= -\frac{45}{8} - \frac{15\kappa}{4} - \frac{41\kappa^2}{48}. \end{aligned} \quad (14.7)$$

The $M = 1$ case proves the OEIS conjecture

$$a_n \sim \frac{13\sqrt{2}}{768} n^{-3/2} \exp\left(\pi\sqrt{\frac{13n}{6}}\right). \quad (14.8)$$

For readers preferring the unshifted variable, the first correction is

$$a_n = Cn^{-3/2}e^{2\sqrt{\kappa}n} \left[1 - \left(\frac{\sqrt{\kappa}}{6} + \frac{3}{2\sqrt{\kappa}} \right) n^{-1/2} + O(n^{-1}) \right]. \quad (14.9)$$

The shift $N = n - 1/6$ keeps the higher formulas shorter.

Remark 14.2 (write, 2 October 2026: consistency with Part I). Take $M = 4$ in Theorem 14.1 and re-expand in the unshifted index. With $\kappa = K^1$ this gives

$$a_n = Cn^{-3/2}e^{2\sqrt{\kappa}n} \left(1 - \frac{d}{\sqrt{n}} + \frac{e}{n} + \frac{f}{n^{3/2}} + O(n^{-2}) \right),$$

with exactly the d, e, f of (1.3)–(1.5): the three symbolic differences vanish (exact SymPy arithmetic, at intake and again in this write). So Theorem 14.1 re-proves Theorem 1.1 with the sharper relative remainder $O(n^{-2})$, by a route independent of Part I's squeeze and finite cut, and continues it to every order. Equation (14.9) is Part I's first correction $-d$, and (14.8) is the leading asymptotic proved in both parts. Part I keeps its own statement and proof.

Literature status and prefactor. The inspected OEIS entry [1] explicitly labels (14.8) conjectural. The published 2023 paper [3] describes its coefficient analysis as conjectural, and the 2024 paper [6, p. 19] repeats that status. The introduction of [3] and its quoted numerical value $0.023938510821419\dots$ agree with C above, while its final display in Section 2 prints the reciprocal. Our derivation fixes the normalization independently. Targeted searches through 1 October 2026 did not locate an earlier proof or an all-orders expansion; this is a scoped search result, not an exhaustive claim of priority.

[write] Added 2 October 2026, batch 77P5. The statement that the final display of Section 2 of [3] prints the reciprocal of C was *not* checked at intake. The shipped mathematical audit (item 13) says the same, but it is the delivery's own review. Part I cites [3] for the conjecture and its equation (1.1) only. As in Part I, the search statement is scoped, and no priority is claimed. Part I's own proof of the leading term and three corrections (Theorem 1.1) predates this manuscript in the repository, though the two were written independently.

15 An exact decomposition of the generating function

All q -series in this section converge for $|q| < 1$. Introduce

$$h_n = \frac{f_{n-1}}{(q; q)_n}, \quad H(z) = \sum_{n \geq 0} h_n z^n \quad (|z| < 1), \quad G(z) = \frac{(-qz^2; q^2)_\infty}{(z; q)_\infty^2}. \quad (15.1)$$

The letter z here is an auxiliary generating-function variable; later $q = e^{-z}$ will be explicitly stated.

Lemma 15.1. *The following identity holds for $|z| < 1$:*

$$H(z) = G(z) \left[1 - \sum_{j \geq 0} \frac{q^j z (q^{j+1} z; q)_\infty^2}{(-q^{2j+1} z^2; q^2)_\infty} \right]. \quad (15.2)$$

Proof. The recurrence gives

$$(1 - z)^2 H(z) = (1 + qz^2) H(qz) - z,$$

whereas G satisfies the corresponding homogeneous equation. Division by G and iteration toward $q^j z \rightarrow 0$ give (15.2). To justify convergence directly, put $\epsilon_k = 2q^k - q^{2k}$. First differences in (14.2) yield

$$f_n = 1 + \sum_{k=1}^n (n - k + 1) \epsilon_k f_{k-2}.$$

Divide by $n + 2$ and apply the discrete Gronwall inequality; the relevant product converges because $\sum k |\epsilon_k| < \infty$. Thus $f_n = O_q(n)$, uniformly on compact q -subdisks. Since $(q; q)_n$ has a nonzero limit, H is analytic for $|z| < 1$. All subsequent rearrangements are absolutely convergent, initially on compact subdisks. $\square$

Define

$$\begin{aligned} P(q) &= \frac{(-q; q^2)_\infty}{(q; q)_\infty^3}, \\ u_l(q) &= \frac{q^l(q^{l+1}; q)_\infty^2}{(-q^{2l+1}; q^2)_\infty}, \\ D(q) &= \sum_{j \geq 0} \frac{(-1)^j q^{j(j-1)/2} (q; q)_j}{(-q; q^2)_{j+1}}. \end{aligned} \quad B(q) = 1 - \sum_{l \geq 0} u_l(q), \quad (15.3)$$

Formula (15.2) gives

$$H(q^{j+1}) = G(q^{j+1}) \left(B + \sum_{l=0}^j u_l \right). \quad (15.4)$$

Proposition 15.2. *There is an exact identity*

$$A(q) = P(q)B(q)D(q) + R(q), \quad (15.5)$$

where $R(q) = 1 + (q; q)_\infty^{-1} E(q)$ and

$$E(q) = \sum_{j \geq 0} \frac{(-1)^j q^{j(j-1)/2}}{(q; q)_j} \sum_{l=0}^j \frac{q^l (q^{l+1}; q)_{j-l}^2}{(-q^{2l+1}; q^2)_{j-l+1}}. \quad (15.6)$$

Proof. Euler's q -binomial theorem gives

$$\frac{1}{(q; q)_{n-1}} = \frac{(q^n; q)_\infty}{(q; q)_\infty} = \frac{1}{(q; q)_\infty} \sum_{j \geq 0} \frac{(-1)^j q^{j(j-1)/2 + nj}}{(q; q)_j}.$$

Substitute this in $A - 1 = \sum_{n \geq 1} q^n h_n / (q; q)_{n-1}$ and interchange sums. The constant piece vanishes because

$$\sum_{j \geq 0} \frac{(-1)^j q^{j(j-1)/2}}{(q; q)_j} = (1; q)_\infty = 0.$$

For the homogeneous part of (15.4), use

$$G(q^{j+1}) = \frac{(-q; q^2)_\infty}{(q; q)_\infty^2} \frac{(q; q)_j^2}{(-q; q^2)_{j+1}}.$$

The remaining product simplifies, for $l \leq j$, to

$$G(q^{j+1}) u_l = \frac{q^l (q^{l+1}; q)_{j-l}^2}{(-q^{2l+1}; q^2)_{j-l+1}},$$

which proves the proposition. $\square$

16 The square identity and a partial theta representation

Proposition 16.1. *One has*

$$B(q) = D(q) = 1 - \sum_{k \geq 0} q^{k(k+1)} \frac{1 - q^{2k+1}}{1 + q^{2k+1}}. \quad (16.1)$$

Consequently

$$A(q) = P(q)B(q)^2 + R(q). \quad (16.2)$$

Proof. Set $a = i\sqrt{q}$, $b = -i\sqrt{q}$, so $a + b = 0$, $ab = q$. Both choices of square root give the same symmetric expressions. We use the convention

$${}_2\phi_1(a, b; c; q, z) = \sum_{n \geq 0} \frac{(a, b; q)_n}{(c, q; q)_n} z^n.$$

For the transformation argument assume $q \neq 0$; the identities at $q = 0$ follow by continuity. Heine's first transformation [10, 17.6.6] applied to ${}_2\phi_1(a, b; q; q, q)$ gives

$$S := 1 - B = \sum_{n \geq 0} \frac{b^n}{1 - aq^n} = \sum_{n, k \geq 0} a^k b^n q^{nk}. \quad (16.3)$$

Indeed, the defining sum of the u_l is

$$\frac{(q; q)_\infty^2}{(a, b; q)_\infty} {}_2\phi_1(a, b; q; q, q).$$

After Heine's transformation, the series ratio is $(a; q)_n / (aq; q)_n = (1 - a) / (1 - aq^n)$; the product prefactor cancels with $1 - a$. Formula (16.3) is symmetric in a, b .

Let $V = (1 + q)D$ and let V_q denote the same summation with an extra factor q^n in term n . Fine's second transformation [10, 17.6.10], in the form

$$(1 - z){}_2\phi_1(q, \alpha q; \beta q; q, z) = \sum_{n \geq 0} \frac{(\beta/\alpha; q)_n (-\alpha z)^n q^{n(n+1)/2}}{(\beta q, zq; q)_n},$$

with $\alpha = a/q^2$, $\beta = a/q$, $z = b$, yields

$$\frac{V - aV_q}{1 - a} = (1 - b){}_2\phi_1(q, a/q; a; q, b) = (1 - b)[1 + (b - 1)S]. \quad (16.4)$$

For the second equality, split off $n = 0$ and use

$$\frac{(a/q; q)_n}{(a; q)_n} = \frac{1 - a/q}{1 - aq^{n-1}}, \quad ab = q.$$

Swap a, b in (16.4). Multiplying the original and swapped equations by $1 - a$ and $1 - b$ and adding gives $2V = 2(1 + q)(1 - S)$. Hence $D = B$.

Finally split the double sum (16.3) into its diagonal and two off-diagonal triangles. On a diagonal difference d , $a^d + b^d$ vanishes for odd d , while it is $2(-1)^m q^m$ for $d = 2m$. Therefore

$$S = \sum_{k \geq 0} q^{k(k+1)} \left(1 + 2 \sum_{m \geq 1} (-1)^m q^{m(2k+1)} \right),$$

and summing the geometric progression gives (16.1). $\square$

Remark 16.2 (write, 2 October 2026: Part I's boundary series is the partial theta function). For $|q| < 1$, Part I's series $D^I(q) = L^I(q)/R^I(q)$ of Section 8.1, with L^I the normalized limiting slope (3.1) of the solution $h_n^I = f_{n-1}$, equals $B(q) = D(q)$. Indeed, in (15.2) the bracket is analytic for $|z| < |q|^{-1/2}$ (the zeros of $(-q^{2j+1}z^2; q^2)_\infty$ have $|z| \geq |q|^{-1/2}$, and the factor q^j makes the sum converge) and equals $B(q)$ at $z = 1$, while G has in $|z| < |q|^{-1}$ only the double pole at $z = 1$, with $(1-z)^2 G(z) \rightarrow (-q; q^2)_\infty / (q; q)_\infty^2$. Hence

$$\frac{f_{n-1}}{(q; q)_n} = [z^n]H(z) = \frac{(-q; q^2)_\infty B(q)}{(q; q)_\infty^2} n + O_q(1),$$

so the slope of Lemma 3.1 is $s(q) = (-q; q^2)_\infty B(q) / (q; q)_\infty$, and $L^I = s / (q; q)_\infty^2 = R^I B$. A 40-digit evaluation at $q = 0.3, 0.6$ and $0.3 + 0.4i$ agrees to 10^{-37} . Consequences: Part I's formal limit $\mathcal{D}(t)$ in (8.2) is the Poincaré expansion of $B(e^{-t})$ given by Lemma 18.1, and the coefficients (1.10) are four times those of $D(e^{-z})^2$ in (18.1); Part I's rational approximants h_m/Q_m in (8.1) approximate this partial theta function; and Theorem 8.1 also follows from (16.2), since $P = R^I$ and R is exponentially smaller on the radius by Lemma 17.1. This answers part of Part I's research question 4 (Section 11): D^I has the closed form (16.1) on the whole unit disk. Its continuation beyond $|q| < 1$ and the zeros of the finite denominators Q_m are not addressed.

17 Global estimates and the exponentially smaller part

These bounds hold on the entire coefficient-extraction circle, not only on the positive real radius. Let $r = e^{-t}$, $t > 0$, and $|q| = r$.

Lemma 17.1. *As $t \downarrow 0$, uniformly on $|q| = r$,*

$$\begin{aligned} |B(q)| &= O\left(t^{-1} \log \frac{1}{t}\right), \\ |R(q)| &= O\left(t^{-1/2} e^{\pi^2/(2t)}\right). \end{aligned} \tag{17.1}$$

Proof. From (16.1),

$$|B(q)| \leq 1 + \sum_{k \geq 0} r^{k(k+1)} \frac{1 + r^{2k+1}}{1 - r^{2k+1}}.$$

Split at $k \asymp t^{-1/2}$ and $k \asymp t^{-1}$. The small- k part is bounded by a harmonic sum times t^{-1} , and the Gaussian tails give the stated estimate.

In (15.6), bound the finite numerator products by $(-r; r)_\infty^2$, the finite denominator products from below by $(r; r^2)_\infty$, and $\sum_{l=0}^j r^l$ by $(1-r)^{-1}$. Euler's identity gives

$$\sum_{j \geq 0} \frac{r^{j(j-1)/2}}{(r; r)_j} = (-1; r)_\infty = 2(-r; r)_\infty.$$

Thus

$$|R(q)| \leq 1 + \frac{2(-r; r)_\infty^3}{(1-r)(r; r)_\infty(r; r^2)_\infty}. \tag{17.2}$$

The standard eta-product estimates

$$(r; r)_\infty \sim \sqrt{\frac{2\pi}{t}} e^{-\pi^2/(6t)},$$

$$(-r; r)_\infty \sim 2^{-1/2} e^{\pi^2/(12t)}, \quad (r; r^2)_\infty \sim \sqrt{2} e^{-\pi^2/(12t)}$$

prove the second bound. Crucially, $\pi^2/2 < \kappa = 13\pi^2/24$. □

Corollary 17.2. *Let $p_n = [q^n]P(q)B(q)^2$. Then*

$$a_n - p_n = O\left(n^{1/4} e^{\pi\sqrt{2n}}\right). \quad (17.3)$$

After Theorem 14.1, this also gives

$$\frac{a_n - p_n}{a_n} = O\left(n^{7/4} e^{-\pi(\sqrt{13/6} - \sqrt{2})\sqrt{n}}\right). \quad (17.4)$$

Proof. Cauchy's estimate and Lemma 17.1 bound the coefficient of R by $O(t^{-1/2} \exp(nt + \pi^2/(2t)))$. Choose its own optimizing radius $t = \pi/\sqrt{2n}$. This is independent of the principal saddle used below. □

This is an exponentially smaller remainder for an *exact principal generating function*. It is not a claim that the algebraic series in Theorem 14.1 converges, nor a complete transseries resolving all smaller exponential scales.

Lemma 17.3. *There are constants $c, t_0 > 0$ such that for $0 < t < t_0$ and $-\pi \leq \theta \leq \pi$,*

$$\frac{|P(e^{-t+i\theta})|}{P(e^{-t})} \leq \exp\left[-c \min\left(\frac{\theta^2}{t^3}, \frac{1}{t}\right)\right]. \quad (17.5)$$

Proof. Each numerator factor satisfies $|1 + q^{2j+1}| \leq 1 + r^{2j+1}$. For denominator factors on the consecutive block $t^{-1} \leq j \leq 2t^{-1}$,

$$|1 - r^j e^{ij\theta}|^2 = (1 - r^j)^2 + 2r^j(1 - \cos j\theta).$$

Since r^j stays in a compact subinterval of $(0, 1)$, taking logarithms gives an upper bound $-c_1 \sum_j (1 - \cos j\theta)$ for the product ratio. For $|\theta| \leq c_2 t$, the quadratic cosine bound gives $c_3 \theta^2/t^3$. For $|\theta|/t$ in a fixed compact interval away from zero, rescaled Riemann sums converge to the strictly positive integral $\int_1^2 (1 - \cos(ux)) dx$. For $|\theta| \geq C_1 t$, choose C_1 large and use $|\sum_j e^{ij\theta}| \leq 1/|\sin(\theta/2)|$ to obtain a lower bound c_4/t . These ranges cover the circle. □

[write] **Added 2 October 2026, batch 77P5.** The proofs of Lemmas 17.1 and 17.3 are sketched at the level of standard estimates (the split of the harmonic sum, and the Riemann-sum compactness step); items 6–8 of the shipped mathematical audit spell them out. Since $P = R^1$, Lemma 17.3 is Part I's bound (6.3) for the same product, proved by the same argument; both texts are kept.

18 Uniform local expansions

From now on put $q = e^{-z}$, with $t = \Re z > 0$ and $|\Im z| \leq t/8$.

Lemma 18.1. *For every fixed M , $D(e^{-z})$ has a uniform Poincaré expansion through degree $M - 1$, obtained by Taylor expanding only its summands $0 \leq j < M$. Its remaining tail is $O(z^M)$.*

Proof. Write T_j for the j th summand of D . Then

$$\frac{T_{j+1}}{T_j} = -q^j \frac{1 - q^{j+1}}{1 + q^{2j+3}}.$$

Assume $0 < t \leq 0.1$ and put $x = (j + 1)t$. If $x \leq 1$, then $|(2j + 3)\Im z| \leq (2 + t)/8 < \pi/2$, so the denominator has real part at least one. Also $|1 - e^{-(j+1)z}| \leq (j + 1)|z|$. Therefore the ratio has modulus at most

$$e^{0.1} \frac{\sqrt{65}}{8} x e^{-x} \leq \frac{e^{0.1} \sqrt{65}}{8e} < \frac{2}{3}.$$

If $x \geq 1$, the bound is

$$e^t \frac{e^{-x}(1 + e^{-x})}{1 - e^{-2x}} = \frac{e^t}{e^x - 1} \leq \frac{e^{0.1}}{e - 1} < \frac{2}{3}.$$

Since $T_M = O(z^M)$ for fixed M , geometric summation proves the uniform tail estimate. $\square$

The first coefficients are

$$\begin{aligned} D(e^{-z}) &= \frac{1}{2} - \frac{z^2}{8} + \frac{z^3}{8} - \frac{11z^4}{48} + \frac{15z^5}{32} - \frac{6737z^6}{5760} + \frac{6451z^7}{1920} - \frac{888521z^8}{80640} + O(z^9), \\ D(e^{-z})^2 &= \frac{1}{4} - \frac{z^2}{8} + \frac{z^3}{8} - \frac{41z^4}{192} + \frac{7z^5}{16} - \frac{6317z^6}{5760} + \frac{1529z^7}{480} - \frac{1702513z^8}{161280} + O(z^9). \end{aligned} \quad (18.1)$$

These are exact rational coefficients, derived from finite Taylor arithmetic.

The eta quotient identity

$$P(q) = \frac{(q^2; q^2)_{\infty}^2}{(q; q)_{\infty}^4 (q^4; q^4)_{\infty}} \quad (18.2)$$

and eta inversion [10, 23.18.5] imply uniformly in the same wedge

$$P(e^{-z}) = \left(\frac{z}{2\pi}\right)^{3/2} \exp\left(\frac{\kappa}{z} - \frac{z}{6}\right) \left(1 + O(e^{-c/t})\right), \quad (18.3)$$

with the principal power of z . For example, the standard product formula is

$$(e^{-z}; e^{-z})_{\infty} = \sqrt{\frac{2\pi}{z}} \exp\left(-\frac{\pi^2}{6z} + \frac{z}{24}\right) (1 + O(e^{-c/t})).$$

Substitution at $z, 2z, 4z$ proves both the constant and the shift in (18.3).

19 Coefficient extraction to every fixed order

Proof of Theorem 14.1. Use Cauchy's coefficient integral on $q = e^{-t+i\theta}$ with $t = \sqrt{\kappa/N}$ and separate the exact remainder R . Lemma 17.1 makes its coefficient exponentially smaller than the result. Let

$$\theta_0 = t^{3/2} \log(1/t).$$

On $|\theta| > \theta_0$, Lemma 17.3, together with the polynomial bound for B , makes PB^2 smaller than every algebraic relative order. On the remaining arc set $z = t - i\theta$. Lemma 18.1 and (18.3) give, for any fixed M ,

$$P(e^{-z})B(e^{-z})^2e^{nz} = (2\pi)^{-3/2}e^{\kappa/z+Nz} \left(\sum_{m=0}^{M-1} b_m z^{3/2+m} + O(|z|^{3/2+M}) \right), \quad (19.1)$$

up to an exponentially small term. The real part of the phase has Gaussian decrease $-c\theta^2/t^3$; hence the integrated remainder is a relative $O(t^M) = O(N^{-M/2})$.

For explicit coefficients, consider a monomial z^α in (19.1), with $\alpha = 3/2 + m$. Deform the truncated vertical saddle segment to the corresponding arc of $|z| = t$, adding radial connectors. Put $\lambda = \sqrt{\kappa N}$. On the circle the phase has real part $2\lambda \cos \phi$. On either connector, writing $|z|/t = s \in [1, \sec \phi]$, it is $\lambda \cos \phi(s + s^{-1})$. At the endpoint angle both are at most $2\lambda - c \log^2(1/t)$. Extending the circle to a Hankel contour cut along the negative real axis therefore changes the integral only by a superalgebraically small relative amount. The negative rays $z = -s$, $s \geq t$, have phase $-\kappa/s - Ns \leq -2\lambda$.

The resulting Hankel integral is

$$\frac{1}{2\pi i} \int_{\mathcal{H}} z^\alpha e^{\kappa/z+Nz} dz = \left(\frac{\kappa}{N} \right)^{(\alpha+1)/2} I_{-\alpha-1}(2\sqrt{\kappa N}). \quad (19.2)$$

One can verify (19.2) by expanding $e^{\kappa/z}$ and applying the reciprocal-Gamma Hankel formula term by term; $|z| \geq t$ ensures convergence. For fixed half-integral order the growing-exponential expansion of I [10, 10.40.1] terminates. Its coefficient at order k is

$$\frac{(-1)^k (m+2+k)!}{(m+2-k)! k! (4\sqrt{\kappa N})^k}, \quad 0 \leq k \leq m+2.$$

Collecting total order $m+k=r$ gives (14.6). The leading constant is

$$\frac{(2\pi)^{-3/2} b_0 \kappa}{2\sqrt{\pi}} = \frac{13\sqrt{2}}{768}, \quad b_0 = \frac{1}{4},$$

as required. □

[write] Added 2 October 2026, batch 77P5. The transfer (19.2)–(14.6) is Part I's (6.4)–(6.5) written in $N = n - 1/6$: with $m = j$, $\beta_k(j)/(2\sqrt{\kappa})^k$ equals the coefficient displayed above, and the factor $\exp(-z/6)$ of (18.3), which Part I expands into its s_j , is absorbed by the shift. Part I applies the transfer to R^1 and to $F_3 R^1$; Part II applies it to PB^2 , whose complement R is exponentially smaller, and that is what yields every order.

Corollary 19.1. *The sequence is eventually strictly log-concave: $a_n^2 > a_{n-1}a_{n+1}$ for all sufficiently large n . More precisely,*

$$\log \frac{a_n^2}{a_{n-1}a_{n+1}} = \frac{\sqrt{\kappa}}{2N^{3/2}} - \frac{3}{2N^2} + O(N^{-5/2}). \quad (19.3)$$

Proof. Take $M = 5$ in Theorem 14.1 and expand the logarithm. For fixed real constants $d_1, \dots, d_4$ this gives

$$\log a_n = g(N) + O(N^{-5/2}), \quad g(u) = \log C + 2\sqrt{\kappa}u - \frac{3}{2} \log u + \sum_{r=1}^4 d_r u^{-r/2}.$$

Taylor expansion of this explicit smooth function gives

$$g(N+1) - 2g(N) + g(N-1) = -\frac{\sqrt{\kappa}}{2N^{3/2}} + \frac{3}{2N^2} + O(N^{-5/2}).$$

The second difference of the three remainder values is still $O(N^{-5/2})$ by the triangle inequality; no derivative bound on the asymptotic remainder is assumed. Negating the resulting second difference proves (19.3), whose leading term is positive. $\square$

[write] Added 2 October 2026, batch 77P5. From the exact coefficients through $n = 2000$: $a_n^2 > a_{n-1}a_{n+1}$ for $3 \leq n \leq 1999$, and it fails only at $n = 1, 2$; the difference between $\log(a_n^2/(a_{n-1}a_{n+1}))$ and the two displayed terms, times $N^{5/2}$, is 0.674 at $n = 500$ and 0.593 at $n = 1999$. These are numerical observations, not part of the proof; no effective threshold is claimed.

20 Continuous inverse models and integer thresholds

All inverse statements below concern the increasing large branch. Put

$$y = 2\sqrt{\kappa(n-1/6)}, \quad K = C(2\sqrt{\kappa})^3, \quad L = \log(X/K), \quad w = \log L. \quad (20.1)$$

The transformed expansion has the form

$$X \sim K \frac{e^y}{y^3} \left(1 + \sum_{r \geq 1} \frac{h_r}{y^r} \right), \quad h_r = (2\sqrt{\kappa})^r c_r. \quad (20.2)$$

Its first coefficients are

$$\begin{aligned} h_1 &= -3, & h_2 &= 3 - 2\kappa^2, \\ h_3 &= 4\kappa^3 + 20\kappa^2, & h_4 &= -\frac{41\kappa^4}{3} - 60\kappa^3 - 90\kappa^2. \end{aligned}$$

If $\log(1 + \sum h_r y^{-r}) \sim \sum \ell_r y^{-r}$, then

$$\begin{aligned} \ell_1 &= -3, & \ell_2 &= -\frac{3}{2} - 2\kappa^2, \\ \ell_3 &= 4\kappa^3 + 14\kappa^2, & \ell_4 &= \frac{9}{4} - 42\kappa^2 - 48\kappa^3 - \frac{47\kappa^4}{3}. \end{aligned} \quad (20.3)$$

20.1 A branch explicit Lambert inverse

The leading equation $X = Ke^{y_0}/y_0^3$ has the exact large solution

$$y_0 = -3W_{-1}\left(-\frac{1}{3}(K/X)^{1/3}\right). \quad (20.4)$$

The branch W_{-1} is essential; it gives $y_0 \rightarrow \infty$. Reverting (20.2) around y_0 gives

$$y = y_0 + \frac{3}{y_0} + \frac{21/2 + 2\kappa^2}{y_0^2} + \frac{45/2 - 8\kappa^2 - 4\kappa^3}{y_0^3} + O(y_0^{-4}), \quad n = \frac{1}{6} + \frac{y^2}{4\kappa}. \quad (20.5)$$

These are inverse asymptotic models, not an assertion that a discrete sequence has a canonical noninteger inverse.

20.2 Logarithmic inverse polynomials

Substitute

$$y = L + 3 \log L + \sum_{j \geq 1} P_j(\log L) L^{-j}$$

into $L = y - 3 \log y + \sum_{r \geq 1} \ell_r y^{-r}$. At each new order P_j enters linearly with coefficient one. This determines every polynomial recursively. In particular,

$$\begin{aligned} y = L + 3w + \frac{9w + 3}{L} + \frac{-27w^2/2 + 18w + 21/2 + 2\kappa^2}{L^2} \\ + \frac{27w^3 - 189w^2/2 - 9w + 45/2 - 12\kappa^2w - 8\kappa^2 - 4\kappa^3}{L^3} + O\left(\frac{w^4}{L^4}\right). \end{aligned} \quad (20.6)$$

After any fixed order J , the remainder in the continuous inverse model for y is

$$O\left(\frac{(\log L)^{J+1}}{L^{J+1}}\right).$$

Squaring and dividing by 4κ converts this to an index-scale error. The Lambert model and the logarithmic polynomials describe the same reversion in different coordinates.

20.3 Shrinking integer brackets before rounding

For $M \geq 1$, define the explicit real function

$$F_M(x) = C(x - 1/6)^{-3/2} e^{2\sqrt{\kappa(x-1/6)}} \sum_{r=0}^{M-1} c_r(x - 1/6)^{-r/2}. \quad (20.7)$$

It is positive and increasing for sufficiently large x , with

$$\frac{d}{dx} \log F_M(x) = \frac{\sqrt{\kappa}}{\sqrt{x}} + O(x^{-1}).$$

Let $x_M(X)$ be its unique large solution of $F_M(x) = X$, and let $n_*(X) = \min\{n : a_n \geq X\}$.

Corollary 20.1. *For each fixed $M \geq 2$, there are constants $D_M, X_M > 0$ such that, for $X \geq X_M$,*

$$\lceil x_M(X) - \delta_M(X) \rceil \leq n_*(X) \leq \lceil x_M(X) + \delta_M(X) \rceil, \quad \delta_M(X) = D_M x_M(X)^{-(M-1)/2}. \quad (20.8)$$

Thus the real bracket width tends to zero. If it does not cross an integer, the threshold is determined exactly by rounding its endpoints.

Proof. Theorem 14.1 gives $a_n/F_M(n) = 1 + O(n^{-M/2})$. The derivative above converts this relative error to an index displacement $O(n^{-(M-1)/2})$. Enlarge its implied constant to obtain strict comparisons on either side of $x_M \pm \delta_M$.

The sequence is nondecreasing: choose the topmost maximum-width row and adjoin one cell immediately to its left outside the bounding box. Row intervals of an L -convex polyomino are nested, since otherwise opposite extreme cells have no internal L path. Thus the chosen row spans all columns, and the new cell connects to every old cell through that row using at most one turn. The enlarged row is uniquely longest, so removing its new leftmost cell recovers the original. This is an injection. Eventual strict increase also follows from the two-term expansion, because the positive order- $n^{-1/2}$ term in a_{n+1}/a_n dominates its order- n^{-1} uncertainty. Monotonicity and the two comparisons prove (20.8). $\square$

The constants in (20.8) are asymptotic existence constants. This report does not supply numerically certified values or a finite starting index. A small numerical model error by itself is therefore not a certified rounding rule, especially at a target $X = a_n$.

[write] Added 2 October 2026, batch 77P5: an instance of repository results. As for Part I's inverse (the note in Section 9.2), this section is an instance of the transseries-and-inversion volume [8], and no novelty is claimed for the method. The core $y_0 - 3 \log y_0 = L$ is `p0:thm:lambert-core` with $a = 1, b = -3$ and the branch W_{-1} ; the corrections in (20.5) and the polynomials (20.6) are the reversion of `p0:thm:perturbed-inversion`, in the Lambert-centred form `p0:thm:lambert-centered`; Corollary 20.1 is part (2), the separation condition, of `p0:thm:staircase`, applied to the model F_M with the model error moved into δ_M .

Relation to Part I's Theorem 9.1. Since $K = c(B^1)^3$, Part II's L at $X = y^1$ is Part I's ℓ , and $y_0 = s_0$. Part II's y is $B^1 \sqrt{n-1/6}$, while Part I's s is $B^1 \sqrt{v}$; hence $s = y + (B^1)^2/(12y) + \dots$, and $y = y_0 + 3/y_0 + \dots$ gives $s = s_0 + D_0/s_0 + \dots$ with $D_0 = (B^1)^2/12 + 3$, Part I's first correction. The two inverses invert different real functions (Part II's model F_M , Part I's interpolation $\tilde{a}$), so they agree to the orders both control, not term by term beyond them. Part II's $n_*(X)$ is Part I's $N(y)$; with monotonicity proved for every n , Part II needs no starting index n_0 in the definition. The shipped data/`02-all-orders-inverse-series.json` also records a fourth Lambert-centred coefficient, $(47\kappa^4/3 + 36\kappa^3 + 45/4)/y_0^4$, and a fourth logarithmic polynomial. They come from formal substitution (residual zero through order 4) and are not printed in the manuscript.

21 Exact checks and reproducible calculations

The proofs above do not infer their constants from numerical fits. For a separate check, set $g_n = f_{n-1}/(q; q)_n^2$. Then

$$(1 - q^n)^2 g_n = 2g_{n-1} - g_{n-2}, \quad g_{-1} = g_0 = 1, \quad A(q) = 1 + \sum_{n \geq 1} q^n (1 - q^n) g_n. \quad (21.1)$$

Truncated power series division by $1 - q^n$ is a simple shifted addition; using it twice gives exact integer coefficients in $O(N^2)$ arithmetic operations up to area N . The replay computes through $N = 2000$ and matches the first 37 displayed OEIS values.

Table 2 gives $(F_M(n)/a_n) - 1$ for selected truncations. More terms improve these examples substantially, but a Poincaré expansion is a fixed-order statement as $n \rightarrow \infty$; indefinitely increasing M at a fixed small n need not improve accuracy. The package contains exact coefficient generation,

n	$M = 1$	$M = 3$	$M = 5$	$M = 9$
100	$8.8963 \cdot 10^{-2}$	$-9.3690 \cdot 10^{-3}$	$-1.7838 \cdot 10^{-3}$	$-5.2974 \cdot 10^{-4}$
200	$5.8496 \cdot 10^{-2}$	$-3.4962 \cdot 10^{-3}$	$-3.4199 \cdot 10^{-4}$	$-2.6476 \cdot 10^{-5}$
500	$3.4317 \cdot 10^{-2}$	$-9.3753 \cdot 10^{-4}$	$-3.7809 \cdot 10^{-5}$	$-4.9192 \cdot 10^{-7}$
1000	$2.3243 \cdot 10^{-2}$	$-3.4290 \cdot 10^{-4}$	$-7.0411 \cdot 10^{-6}$	$-2.3623 \cdot 10^{-8}$
2000	$1.5900 \cdot 10^{-2}$	$-1.2444 \cdot 10^{-4}$	$-1.2961 \cdot 10^{-6}$	$-1.1150 \cdot 10^{-9}$

Table 2: Signed relative error of the shifted forward model.

rational expansion arithmetic, inverse reversion checks, and independent evaluations of the exact decomposition at real and nonreal points. These checks test transcription and normalization; the complex bounds and coefficient-transfer proof supply the asymptotic theorem. A separate audit records the precise source version reviewed. Neither these computational checks nor that review constitute machine-checked formal verification or external peer review.

[write] Added 2 October 2026, batch 77P5: files and rerun. In the repository the replay is `code/02-all-orders-replay.py` (delivered as `code/replay.py`); its recorded outputs are the files `data/02-all-orders-*.json`, and the “separate audit” is the three `02-all-orders-*.md` reviews. The replay reads its reference fixture from `../data/coefficients-2000.txt` relative to itself, a file shipped as `data/02-all-orders-coefficients-2000.txt`, so it runs only in the delivered layout; the report’s README recreates that layout on a scratch copy. The fixture agrees entry by entry with the column a_n of Part I’s `data/coefficients.csv`, $0 \leq n \leq 2000$: the two parts computed the same 2001 integers by different recurrences. On 2 October 2026 that recipe passed here: the core replay in 6 s, and the replay with the optional SymPy and mpmath parts in 78 s. All five outputs agreed with the shipped records apart from line endings (Windows writes CRLF).

22 What remains open

1. **Effective constants and thresholds.** Quantifying all remainder bounds would turn the shrinking asymptotic brackets into certified finite algorithms with explicit starting ranges.
2. **Smaller exponential scales.** The exact remainder bound (17.3) is deliberately coarse. Determining the actual dominant singular behavior of R , and then secondary contributions from PB^2 , would be needed for a complete exponentially improved expansion.
3. **Large correction order.** The growth of the b_m and c_r , optimal truncation, and possible resummation are separate problems. The fixed-order proof does not settle them.
4. **Further convexity classes.** The decisive step here is the exact product-square decomposition. Whether comparable transformations can be derived from area generating functions for other constrained polyomino classes is a concrete direction; this result alone makes no assertion about their asymptotics.
5. **The partial theta factor.** The Lambert, partial theta, and rapidly convergent q -series forms of B may permit sharper arithmetic or analytic descriptions. No modularity claim for B is required here.

[write] Added 2 October 2026, batch 77P5. These questions overlap Part I’s research questions of Section 11: item 1 continues Part I’s question 2 (explicit error constants), item 2 its question 5 (beyond-all-orders terms), item 4 its question 7 (Z-convex and other classes), and item 5 its question 4, since B is Part I’s D^I (Remark 16.2). Part I’s question 1 is answered by Theorem 14.1.

23 Provenance of the two parts, and where the merge had to choose

This section was written when Part II was added.

Two manuscripts. The report is built from two manuscripts, each printed in full.

- **Part I** (Sections 1–12, Appendices A and B): manuscript 05 of batch 75, “L-convex polyominoes and a colored-partition comparison”, archive A126764_Lconvex_Research.zip, arrival commit 4b874cea0, placed in 6ea60e367 and written in 936896776. It contributed the squeeze, the finite-cut proof of three corrections, the radial theorem, the interpolated inverse and the exact and numerical checks of Section 10. It names no ProveIt commit (its repository input was an overview and a targeted search), so it has no pin.
- **Part II** (Sections 14–22): manuscript 31 of batch 77, “All order asymptotics and inversion for L convex polyominoes”, archive lconvex-asymptotics-reproducibility.zip, arrival commit 096ee7b87, placed in 4f11bc9c0. It contributed the exact product-square decomposition, the all-orders coefficient theorem, log-concavity, monotonicity by injection, and the shifted inverse with integer brackets. It names no ProveIt commit and does not mention the repository, so it has no pin either.

Where the merge had to choose.

- *Placement.* Part II is printed after Part I’s closing Section 12 and before Part I’s appendices, so no section, theorem or equation number of Part I moves. Part I’s Appendix A remains the reproduction appendix of Part I; Part II’s reproduction notes are in Section 21.
- *Letters.* Part II keeps the manuscript’s letters, which the shipped data and reviews use; the clashes with Part I are resolved by Table 1 and by the superscript I in write notes, not by renaming (Section 13.3). The manuscript’s macros `\e`, `\ii` and `\dd` were mapped to Part I’s `\ee` and two new preamble macros; three unused macros of the manuscript were dropped.
- *Duplicated results.* Results proved in both manuscripts keep both proofs, with cross-references (Section 13); no proof was shortened.
- *Bibliography.* The manuscript’s entries for Castiglione–Frosini–Munarini–Restivo–Rinaldi, Guttmann–Kotěšovec, Guttmann–Massazza and OEIS A126764 cite the same works as Part I’s entries [2, 3, 6, 1] and were merged into them (the page reference for Guttmann–Massazza is noted in that entry). Its DLMF entry cites other sections and a stated version, so it is kept as a separate entry [10].
- *Additions.* Section 13, this section, the remarks headed “write” and the [write] notes in Part II are editorial; Remark 16.2 is a write-phase observation, with its proof and a numerical check. No statement, proof or number of either manuscript was changed; labels of Part II carry the prefix `lcp:ao:`.

- *Not shipped.* The delivered PDF of manuscript 31, its build script, its delivery README, its checksum manifest `MANIFEST.json` with the package verifier, the replay's own copy of its coefficient output (equal to the shipped fixture plus a final newline) and the record of its PDF build. The archive survives in the arrival commit; retrieve it with `git show 096ee7b87:docs/incoming/lconvex-asymptotics-reproducibility.zip`.

A Reproduction commands and files

The archive contains this standalone $\text{\LaTeX}$ source, the compiled PDF, two verification scripts, exact coefficient data, and their verification logs. Build the document with

```
pdflatex -interaction=nonstopmode -halt-on-error lconvex_asymptotics.tex
pdflatex -interaction=nonstopmode -halt-on-error lconvex_asymptotics.tex
```

Run the checks from the archive directory with

```
python verify.py --nmax 2000 --out checks
python verify_symbolic.py
```

The first command requires only the Python standard library. The second additionally requires SymPy and mpmath. The quadratic number of arithmetic operations in the coefficient checker is not a quadratic bit-complexity claim: the integer sizes grow with n . The code uses $O(N)$ large-integer storage for each of a fixed number of truncated series. No external data download is required for reproduction.

[write] **Added 2 October 2026, batch 75.** In the repository the source is `article.tex`, the scripts are in `code/`, and the contents of the delivered `checks/` directory are in `data/`; the compiled PDF of the archive is not shipped. `verify.py` writes into `-out` relative to the working directory, while `verify_symbolic.py` reads `coefficients.csv` from, and writes into, a `checks/` directory beside itself. The commands above therefore work only in the delivered layout; the report's README recreates it on a scratch copy. On 2 October 2026 that recipe reproduced `coefficients.csv` and both tables byte for byte and the other outputs apart from line endings.

B A short review checklist

A reader auditing the proof can concentrate on four potential failure points. First, retain the slope term when $z \rightarrow 1$ in (3.4); setting $(1-z)^2 C(z)$ to zero would give a false identity. Second, distinguish coefficientwise positivity from positivity on the real interval; the proof of (4.3) establishes the stronger statements explicitly. Third, the finite-cut error is estimated on the *whole* coefficient circle using positive tails and the controlled full-circle integral of Q_3^{-2} . Fourth, the real-axis all-orders series is not used as a substitute for this complex estimate. These checks isolate the substantive logical steps of the argument.

References

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